



Chapter # 10

Biotechnology



Student Learning Outcomes

After studying this chapter, students will be able to:

- [B-12-J-01] Introduce genetic engineering.
- [B-12-J-02] Explain polymerase chain reaction (PCR).
- [B-12-J-03] Outline the function of restriction enzymes.
- [B-12-J-04] Describe plasmid as vector prokaryotes and explain how recombinant plasmids can be formed.
- [B-12-J-05] Define genetically modified organism.
- [B-12-J-06] Explain the formation of human insulin protein in bacteria.
- [B-12-J-07] Describe how vertical food farms (soil free) work.
- [B-12-J-08] Compare and contrast the advantages of vertical food farms with general agricultural practices prevalent in Pakistan.

Biotechnology is an innovative field that bridges biology and technology, focusing on the use of living organisms, cells, and biological systems to develop products and solutions that improve human life and the environment. At its core, biotechnology is about understanding the molecular and cellular mechanisms of life and applying this knowledge in practical, often transformative ways. Unlike traditional practices that rely on observation or trial and error, biotechnology employs precise scientific methods to manipulate biological components with specific objectives.

The history of biotechnology dates back to the earliest stages of human civilization. Even in ancient times, societies applied basic biotechnological principles when they learned to cultivate crops and domesticate animals, ensuring a more stable and continuous food supply. These early practices, although unrecognized at the time as biotechnology, laid the groundwork for the sophisticated techniques used today. The formal term “biotechnology” was introduced in the late 20th century and is defined as the use of living organisms—particularly microorganisms—or their biological processes and products for the benefit of humanity.

Historically, humans have unknowingly used biotechnological methods in everyday life. For example, fermented foods such as yogurt and cheese were produced from milk using microbial activity, while alcoholic beverages like wine and beer were derived from the fermentation of fruit juices, sugarcane, or malted barley. Even leather tanning involved specific bacterial action, demonstrating an early understanding of microbial utility. These ancient applications established the foundation for modern biotechnology, which now spans agriculture, medicine, industry, and environmental management.

Modern biotechnology builds upon these historical principles, relying heavily on genetics and molecular biology. By studying genes, proteins, and cellular pathways, scientists can identify the mechanisms underlying diseases, crop traits, and industrial processes. This understanding enables targeted interventions, such as modifying the genetic makeup of organisms to express desired traits, producing therapeutic proteins, or designing microbes capable of degrading pollutants. Working at the molecular level has led to innovations that were unimaginable just a few decades ago.

Applications of biotechnology are wide-ranging. In healthcare, it has revolutionized medicine through the development of vaccines, antibiotics, monoclonal antibodies, and personalized therapies. Recombinant DNA technology and gene editing enable the production of insulin, growth hormones, and other essential drugs more efficiently and safely than ever before. In agriculture, genetically modified crops resistant to pests, diseases, or harsh environmental conditions enhance food security and sustainability. Industrial biotechnology harnesses microorganisms and enzymes to produce biofuels, biodegradable plastics, and other eco-friendly materials.



Environmental biotechnology uses living systems to treat wastewater, clean up oil spills, and manage waste through processes such as bioremediation.

Despite its tremendous potential, biotechnology also raises ethical, social, and safety concerns. Manipulation of genetic material, creation of genetically modified organisms (GMOs), and use of stem cells or cloning techniques require careful regulation. Responsible application is essential to balance innovation with public welfare.

In conclusion, biotechnology is a dynamic and interdisciplinary field that harnesses the principles of biology to create solutions for human and environmental needs. From its earliest use in food production and leather tanning to modern genetic engineering and industrial applications, biotechnology has continuously evolved to transform medicine, agriculture, industry, and environmental management. By combining historical knowledge with modern scientific advancements, biotechnology continues to improve health, sustain resources, and address pressing global challenges, making it one of the most exciting and impactful areas of modern science.

SLO [B-12-J-01]

Introduce genetic engineering.

GENETIC ENGINEERING

Genetic engineering is a transformative branch of biotechnology that allows scientists to deliberately modify the genetic material of an organism to alter its characteristics for the benefit of humanity. While biotechnology as a field is vast and multidisciplinary, genetic engineering stands out for its precision and potential to revolutionize medicine, agriculture, industry, and environmental management.

Why Genetic Engineering is Needed

The rapid growth of the global human population, now exceeding 8 billion, has put unprecedented pressure on food production, healthcare, industry, and the environment. Traditional methods of agriculture, medicine, and industrial manufacturing often struggle to keep pace with these rising demands. Crops can fail due to pests, diseases, or extreme weather conditions; livestock may be susceptible to infections; and industrial processes can be resource-intensive and environmentally damaging. In this context, genetic engineering emerges as a powerful tool to provide sustainable, efficient, and targeted solutions.

- **Enhancing Living Organisms:** Genetic engineering allows scientists to directly modify the genetic makeup of organisms to introduce desirable traits. For example, crops can be engineered to tolerate drought, salinity, or high temperatures, ensuring stable yields even in harsh climates. Similarly, livestock can be enhanced to exhibit faster growth, disease resistance, or improved milk and meat quality, reducing losses and increasing productivity. These enhancements make food systems more resilient, reliable, and capable of feeding a growing population.
- **Improving Nutritional Quality and Resistance:** Another critical application is the development of nutritionally enriched and pest-resistant crops. Staple crops like rice, wheat, and maize can be fortified with essential vitamins and minerals, addressing malnutrition in vulnerable populations. At the same time, genes that confer resistance to insects or pathogens can be introduced, reducing the need for chemical pesticides and minimizing environmental damage. For instance, **Bt cotton** and **Bt maize** produce a natural insecticidal protein that protects crops from pest attacks, demonstrating how genetic engineering can improve both yield and sustainability.
- **Advancing Healthcare and Industrial Applications:** Beyond agriculture, genetic engineering enables the creation of microorganisms capable of producing life-saving medicines, such as insulin, vaccines, and antibiotics. Industrial biotechnology also benefits, with engineered microbes producing biofuels, biodegradable plastics, enzymes, and other eco-friendly products, reducing dependence on non-renewable resources and minimizing pollution. This capability makes industrial processes more sustainable while providing environmentally conscious solutions to global challenges.
- **Sustainable Solutions for the Future:** By integrating genetic engineering into agriculture, healthcare, and industry, humanity can achieve efficient, high-yield, and environmentally sustainable solutions. This technology not only addresses immediate needs, such as food security and medical treatment, but also lays the groundwork for long-term sustainability, helping societies adapt to climate change, resource limitations, and emerging global health challenges.



In essence, genetic engineering is not just a scientific advancement—it is a **strategic tool** for meeting the demands of the 21st century, offering **precision, efficiency, and sustainability** in ways traditional methods cannot match. Its ability to enhance organisms, improve nutrition, protect ecosystems, and innovate industrial processes makes it an indispensable component of modern biotechnology.

Genetic Engineering and Recombinant DNA Technology

Genetic engineering is the deliberate modification of an organism's genetic material to change its characteristics in a controlled manner. The practice is largely based on **Recombinant DNA (rDNA) Technology**, which involves combining DNA from **different sources** to form novel genetic combinations.

Recombinant DNA (rDNA) Technology forms the molecular basis of genetic engineering. It involves a series of **in vivo steps** to isolate, combine, and express DNA segments from different sources. This allows scientists to **produce multiple copies of a desired gene** and synthesize its corresponding protein on an industrial scale.

Core Components of Recombinant DNA Technology

1. **Gene of Interest** – The DNA fragment to be cloned and expressed. It can be obtained through:
 - **Artificial Gene Synthesis** – Chemical synthesis in vitro without a DNA template.
 - **Reverse Transcription of mRNA** – Using reverse transcriptase to create a complementary DNA (cDNA).
 - **Direct Isolation from Genomic DNA** – Using **restriction endonucleases** to excise specific genes.
2. **Molecular Scissors (Restriction Endonucleases)** – Enzymes that cut DNA at specific sequences.
3. **Molecular Carriers (Vectors)** – DNA molecules, such as plasmids, used to deliver the gene of interest into host cells.
4. **Molecular Glue (DNA Ligase)** – Enzyme that joins DNA fragments together.
5. **Expression System (Host Cell)** – The organism in which the gene is expressed to produce the desired protein or trait.

Approaches in Genetic Engineering

Genetic engineering provides scientists with **precise tools** to manipulate the DNA of organisms, allowing them to **enhance traits, correct defects, or introduce entirely new capabilities**. Unlike traditional breeding methods, which rely on random genetic recombination over multiple generations, genetic engineering allows for **targeted, predictable, and rapid modifications** at the molecular level. The main approaches used in genetic engineering include:

1. Addition of New Genes

One of the primary approaches is to **introduce new genes** into an organism's genome to confer desirable traits. This is also known as **gene insertion** or **gene augmentation**. For example:

- Crops like **Bt cotton** or **Bt maize** are engineered by adding genes from the bacterium *Bacillus thuringiensis*, which produce proteins toxic to specific insect pests. This makes the plants **naturally pest-resistant**, reducing the need for chemical pesticides.
- In medicine, bacteria can be engineered to carry the **human insulin gene**, enabling large-scale production of insulin for diabetes treatment.

By adding genes from other species or even synthetic genes, scientists can **introduce entirely new functions** that the organism did not naturally possess.

2. Removal of Unwanted Genes

Genetic engineering also allows for the **removal or silencing of genes** responsible for harmful traits, a process called **gene knockout** or **gene silencing**. Examples include:

- Eliminating genes in crops that make them susceptible to viral or fungal infections, resulting in **disease-resistant varieties**.
- Deactivating genes in livestock that contribute to inherited disorders, improving **animal health and productivity**.

This approach not only prevents undesirable traits from being expressed but also contributes to higher yield, better quality, and reduced economic losses.

3. Editing Existing Genes

The third major approach is **gene editing**, which involves **modifying existing genes** to enhance or alter their function without necessarily adding or removing them. Advanced tools like **CRISPR-Cas9** allow scientists to:

- **Correct genetic mutations** responsible for inherited diseases in humans.



- Improve drought tolerance, nutrient content, or growth rates in crops.
- Fine-tune industrial microbes to produce higher quantities of enzymes, biofuels, or biodegradable products.

Gene editing provides **precision at the nucleotide level**, making it possible to make **subtle, highly targeted changes** that were impossible with earlier techniques.

Goals of Genetic Engineering

Genetic engineering is a **purpose-driven science** that aims to **manipulate the genetic material of organisms** to achieve specific, beneficial outcomes. By directly targeting DNA, scientists can **design organisms** to meet human needs in **agriculture, medicine, industry, and environmental management**. The primary goals of genetic engineering include:

1. Enhancing Existing Traits

One major objective is to improve traits that organisms already possess. For example:

- Crops can be engineered to produce **higher yields**, grow faster, or resist environmental stresses like drought, salinity, or extreme temperatures.
- Livestock can be enhanced for **better growth, milk production, or disease resistance**, increasing productivity and sustainability.

By refining existing traits, genetic engineering ensures that organisms perform more efficiently, helping meet the demands of a growing population.

2. Introducing New Capabilities

Another key goal is to **add entirely new traits or functions** to an organism. This can include:

- Introducing genes that provide resistance to pests, viruses, or harmful microorganisms in plants and animals.
- Enriching crops with essential nutrients, such as Vitamin A in Golden Rice, to combat malnutrition.
- Equipping microorganisms to synthesize medicines, enzymes, or biofuels that they could not naturally produce.

This ability to introduce new capabilities opens up solutions that were previously impossible with traditional breeding or chemical methods.

3. Eliminating Harmful Characteristics

Genetic engineering also targets traits that are **detrimental to the organism, humans, or the environment**. Examples include:

- Removing genes that make plants susceptible to **fungal infections or insect attacks**, reducing crop loss.
- Correcting or silencing **mutated genes in humans** responsible for genetic disorders.
- Engineering livestock to minimize **hereditary diseases**, improving animal health and productivity.

Eliminating harmful traits not only protects organisms but also ensures safer, higher-quality food and resources for human use.

4. Engineering Microorganisms for Human Benefit

Microbes are among the most versatile organisms used in genetic engineering. Scientists can:

- Produce **life-saving medicines**, such as insulin, vaccines, and antibiotics.
- Perform **bioremediation**, breaking down pollutants in water, soil, or industrial waste.
- Manufacture **biofuels, biodegradable plastics, and industrial enzymes**, providing environmentally sustainable alternatives to conventional products.

By harnessing microorganisms, genetic engineering creates tools for **health, industry, and environmental sustainability**, making it a cornerstone of modern biotechnology.

Major Applications of Genetic Engineering

1. Agriculture

- Development of **genetically modified (GM) crops** resistant to pests, diseases, drought, or salinity.
- Improvement in **crop yield and nutritional content**, ensuring better food security.

2. Medicine

- **Gene therapy** to treat inherited genetic disorders.
- Production of **biopharmaceuticals** including insulin, vaccines, and antibiotics.
- Exploration of **novel treatments** for diseases like cancer, hepatitis, and other life-threatening conditions.



3. Industry and Environment

- Engineering microorganisms to produce biofuels, biodegradable plastics, and other eco-friendly materials.
- Bioremediation, using microbes to degrade industrial pollutants and sewage.
- Synthesis of environmentally safe industrial chemicals.

A Promising Future:

Genetic engineering is one of the **fastest advancing areas of science**. Dr. Craig Venter, a pioneer in synthetic biology, famously said, "*The 21st century is the Century of Biology.*" In Pakistan, biotechnology has been recognized as a **priority area by the National Commission on Science and Technology**, highlighting its importance for national progress. With ongoing research, genetic engineering promises **far-reaching benefits** for human health, food security, industry, and environmental sustainability.

SLO [B-12-J-03]

Outline the function of restriction enzymes.

RESTRICTION ENZYMES: MOLECULAR SCISSORS OF GENETIC ENGINEERING

Restriction enzymes, also called restriction endonucleases, are specialized proteins that cleave the phosphodiester bonds of both DNA strands at specific nucleotide sequences known as restriction sites. Originally discovered in bacteria, these enzymes serve as a natural defense mechanism against invading viruses (bacteriophages) by "restricting" viral replication.

Function in Nature

In bacterial cells, restriction enzymes **digest foreign DNA**, preventing viral infection. To avoid cutting their own genomic DNA, bacteria **protect themselves through methylation**, where methyl groups are added to nucleotides in their recognition sequences by **methyltransferase enzymes**. This selective protection allows the enzymes to target only foreign DNA while leaving the host genome intact.

Naming Convention

Restriction enzymes are named after the genus, species, and strain of the bacteria from which they were isolated:

- **EcoRI** – Escherichia coli
- **HindII/HindIII** – Haemophilus influenzae
- **XhoI** – Xanthomonas holcicola

Recognition Sequences

Restriction sites are usually **palindromic sequences** of 4–8 base pairs, meaning they read the same backward and forward on complementary DNA strands. For example, EcoRI recognizes **GAATTC** and cuts between G and A. The recognition of these specific sequences ensures **precision in DNA manipulation**.

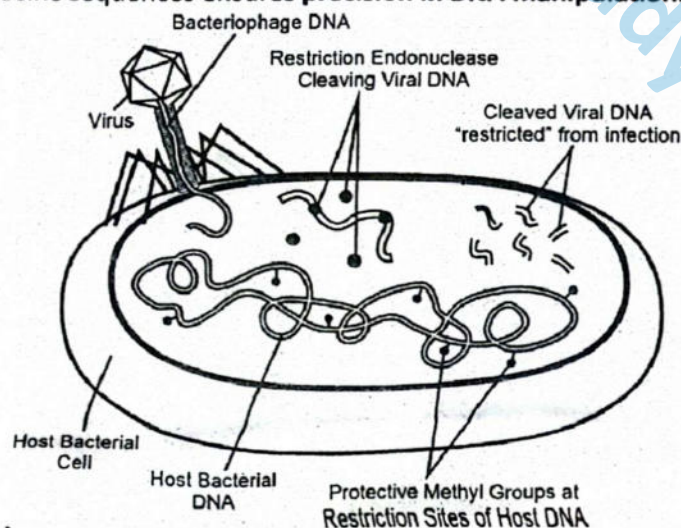


Figure: Functioning of Restriction Endonucleases in Bacterial Cell

Types of Cuts by Restriction Enzymes

Restriction enzymes cut DNA in precise ways, producing fragments that can later be manipulated in genetic engineering. The two main types of cuts are **sticky ends** and **blunt ends**. Understanding these types is crucial because they determine how DNA fragments can be joined, cloned, or expressed in host organisms.

1. Sticky Ends (Staggered Cuts)

Sticky ends, also called **cohesive ends**, are produced when restriction enzymes cut DNA **staggeredly**, leaving **short single-stranded overhangs** at each end of the fragment. These overhangs are complementary to each other, allowing **base pairing with matching sequences** from other DNA fragments.

Key Features and Importance:

- The overhangs can be **5' or 3' single-stranded sequences**, depending on the enzyme. For example:
 - **EcoRI** produces a 5' overhang.
 - **KpnI** produces a 3' overhang.
- Sticky ends facilitate the precise joining of DNA fragments from different sources, which is essential in creating recombinant DNA molecules.
- They allow **efficient ligation** using the enzyme **DNA ligase**, ensuring that the fragments are joined in the correct orientation and sequence.
- Sticky ends are especially useful when **combining genes from different organisms** or inserting a gene into a plasmid vector for cloning.

Example Applications:

- In medicine, sticky ends help insert the **human insulin gene** into **bacterial plasmids**, enabling bacteria to produce insulin efficiently.
- In agriculture, genes conferring **pest resistance** or enhanced nutrition can be inserted into plant genomes using sticky-ended DNA fragments.

2. Blunt Ends

Blunt ends are produced when restriction enzymes cut **straight across both DNA strands**, leaving **no overhangs**. The cut ends are flush, meaning both strands terminate at the same nucleotide position.

Key Features and Importance:

- Blunt-end fragments do not require complementary overhangs for ligation.
- They are versatile because any blunt-ended fragment can, in principle, be joined with any other blunt-ended fragment.
- However, ligation of blunt ends is generally less efficient than sticky ends because there is no complementary base pairing to guide the process.

Example:

- **HaeIII** produces blunt-ended DNA fragments used in genome mapping, DNA sequencing, and other molecular cloning techniques where precise sticky-end pairing is not necessary.

Comparison and Significance in Genetic Engineering

Feature	Sticky Ends	Blunt Ends
Overhang	Yes (single-stranded)	No (flush ends)
Ligation Efficiency	High, guided by complementary sequences	Lower, no guiding overhang
Precision in Recombinant DNA	High	Moderate
Common Enzymes	EcoRI, KpnI	HaeIII
Applications	Gene cloning, plasmid insertion	DNA assembly, blunt-end cloning

CRITICAL THINKING

- If restriction endonuclease enzymes are capable of destroying the DNA of bacteriophage, then why they do not break down the genomic DNA of host cell i.e. bacteria?
- The restriction enzymes that form sticky ends are more useful in genetic engineering. Why? Isolated. For example, **EcoRI** was isolated from *Escherichia coli*, **Hind-II** and **Hind-III** from *Haemophilus influenzae* and **XhoI** from *Xanthomonas holcicola*.



Sticky ends are particularly valuable in **genetic engineering** because they allow **efficient, accurate, and directional joining** of DNA fragments from different organisms. Blunt ends, though less efficient, offer **flexibility** when overhangs are unnecessary. By selecting the appropriate type of cut, scientists can **manipulate DNA with precision**, creating recombinant molecules for medicine, agriculture, and industrial biotechnology.

Applications of Restriction Enzymes in Genetic Engineering

Restriction enzymes are **essential tools in recombinant DNA (rDNA) technology**. By acting as precise molecular scissors, they allow scientists to **manipulate DNA in controlled and targeted ways**, enabling a wide range of applications in medicine, agriculture, and industry. Their ability to cut DNA at **specific sequences** makes them indispensable for **gene cloning, expression, and modification**.

1. Isolating Genes of Interest

Restriction enzymes allow scientists to **cut out specific genes** from the genome of an organism. This is the first step in creating **recombinant DNA molecules**, as the gene of interest can then be studied, cloned, or expressed in another organism. Isolation of genes also enables the **study of gene function**, identification of disease-causing mutations, and understanding of regulatory sequences in DNA.

Example: The human **insulin gene** is isolated using restriction enzymes before being inserted into a bacterial plasmid for mass production.

2. Inserting Genes into Vectors

Once a gene of interest is isolated, it must be **inserted into a DNA vector**—such as a plasmid, bacteriophage, or artificial chromosome—to be replicated and expressed in a host organism. Restriction enzymes generate **compatible sticky or blunt ends** in both the gene and the vector, allowing precise insertion using **DNA ligase**. Vectors such as **pBR322 plasmid** contain multiple restriction sites and selectable markers, which help in **efficient gene insertion and selection of transformed cells**.

Example: In genetically modified crops, a gene conferring **pest resistance** can be inserted into a plant plasmid vector, which is then introduced into plant cells to create a **transgenic plant**.

3. Production of Medicines, Vaccines, and GM Crops

Restriction enzymes enable the creation of recombinant organisms capable of producing **therapeutic proteins, vaccines, and industrial enzymes**.

- **Medicines:** Human insulin, growth hormones, clotting factors, and monoclonal antibodies are produced by bacteria or yeast engineered using **recombinant DNA technology**.
- **Vaccines:** Genes encoding viral proteins can be inserted into vectors to produce safe and effective vaccines, such as the **hepatitis B vaccine**.
- **Genetically Modified (GM) Crops:** Crops like **Bt cotton, Bt maize, and Golden Rice** are created using restriction enzymes to insert genes that confer **pest resistance, enhanced nutrition, or stress tolerance**.

4. Other Research and Industrial Applications

Restriction enzymes are also used in **DNA mapping, genome sequencing, and forensic analysis**. In industry, they allow for the **engineering of microbes** that produce **biofuels, biodegradable plastics, and industrial enzymes**, supporting sustainable and environmentally friendly solutions. These enzymes help scientists **study gene regulation**, understand genetic disorders, and develop **novel biotechnological tools** for research and innovation.

In essence, restriction enzymes form the foundation of genetic engineering. By enabling the isolation, insertion, and recombination of genes, they allow scientists to design organisms with desired traits, produce life-saving medicines and vaccines, and develop genetically modified crops for a sustainable future. Their precision, reliability, and versatility make them indispensable in modern biotechnology, opening the door to innovative solutions across medicine, agriculture, and industry.

Vectors: Molecular Carriers in Genetic Engineering

After DNA is cut using **restriction enzymes**, the next critical step in **genetic engineering** is to insert the gene of interest into a **vector**. Vectors are DNA molecules that serve as **transport vehicles**, carrying recombinant DNA into **host cells** where it can be **replicated and expressed**. They are essential tools in recombinant DNA technology, enabling the production of proteins, enzymes, and genetically modified organisms.

Plasmids: The Most Common Vectors

Plasmids are small, circular, double-stranded DNA molecules naturally found in bacteria. They are particularly suited as vectors because they **replicate independently of the bacterial chromosome** and can carry additional genes. Scientists often engineer plasmids to carry foreign genes for research, medicine, or agriculture.



► Key Features of Plasmid Vectors

1. Origin of Replication (Ori):

- Allows plasmids to **replicate independently** within the host cell.
- Ensures that multiple copies of the recombinant plasmid are present in each cell, increasing gene expression efficiency.

2. Selectable Markers:

- Genes such as **antibiotic resistance markers** (e.g., ampicillin or tetracycline) allow researchers to identify cells that have successfully **taken up the plasmid**.

3. Multiple Cloning Sites (MCS):

- Short sequences containing **unique restriction enzyme sites**, enabling easy insertion of foreign DNA.
- Allow scientists to **cut and insert genes precisely** without disrupting other plasmid functions.

4. Promoter Regions:

- Sequences that **initiate transcription** of the inserted gene, enabling the host cell to produce the desired protein.

5. Small Size:

- Facilitates **manipulation in the laboratory**, as smaller plasmids are easier to isolate, cut, and recombine.

6. High Copy Number:

- Some plasmids can exist in **multiple copies per cell**, ensuring **efficient amplification** of the inserted gene.

► Example: pBR322 Plasmid

- Developed in **1977 by Bolivar and Rodriguez**, pBR322 is one of the earliest and most widely used plasmid vectors.
- **Origin of replication:** Derived from pMB1.
- **Size:** 4,361 base pairs, allowing **efficient transformation** into bacterial cells.
- **Antibiotic resistance genes:** Ampicillin (Amp^R) and tetracycline (Tet^R) for selection.
- **Multiple restriction sites:** PstI, PvuI, SacI (Amp^R); BamHI, HindIII (Tet^R), enabling precise gene insertion.

► Other Types of Vectors

While plasmids are the most common, other vectors are used depending on the application:

- **Bacteriophage λ DNA:** A viral vector that can carry larger DNA fragments than plasmids.
- **Cosmids:** Hybrids of plasmid and phage DNA, allowing the cloning of very large DNA fragments.
- **Yeast Artificial Chromosomes (YACs):** Used to clone extremely large DNA fragments in **yeast cells**, ideal for genomic studies.
- **Bacterial Artificial Chromosomes (BACs):** Can carry large DNA fragments in **bacterial cells**, used in genome sequencing projects.

► Integration with Restriction Enzymes

Restriction enzymes and vectors work hand in hand in genetic engineering:

- **Restriction enzymes** act as **molecular scissors**, cutting DNA at precise palindromic sequences and generating **sticky or blunt ends**.
- These cut fragments are then **inserted into vectors** like plasmids using DNA ligase.
- The recombinant DNA can be introduced into host cells, which will **replicate and express the inserted gene**.

This combination allows scientists to produce medicines (e.g., insulin), genetically modified crops (e.g., Bt cotton), industrial enzymes, and environmentally friendly products, highlighting the central role of vectors and restriction enzymes in modern biotechnology.

SCIENCE TITBITS

Bacteria methylate their DNA shortly after synthesis during binary fission, by adding methyl (CH₃) groups to the nucleotides in the recognition sequences. This process is called methylation and is carried out by the modifying enzyme methyl-transferase present in the bacterial cell.



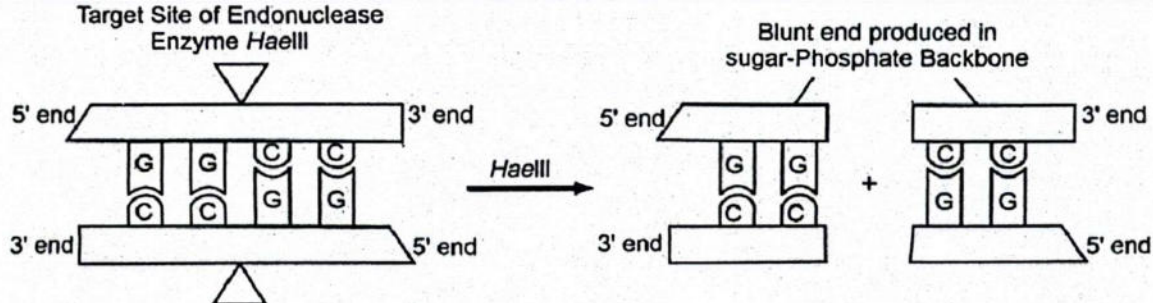


Figure: Recognition of palindromic sequence by restriction endonuclease enzyme *HaeIII* and production of blunt ends in Sugar-Phosphate backbone of double stranded DNA

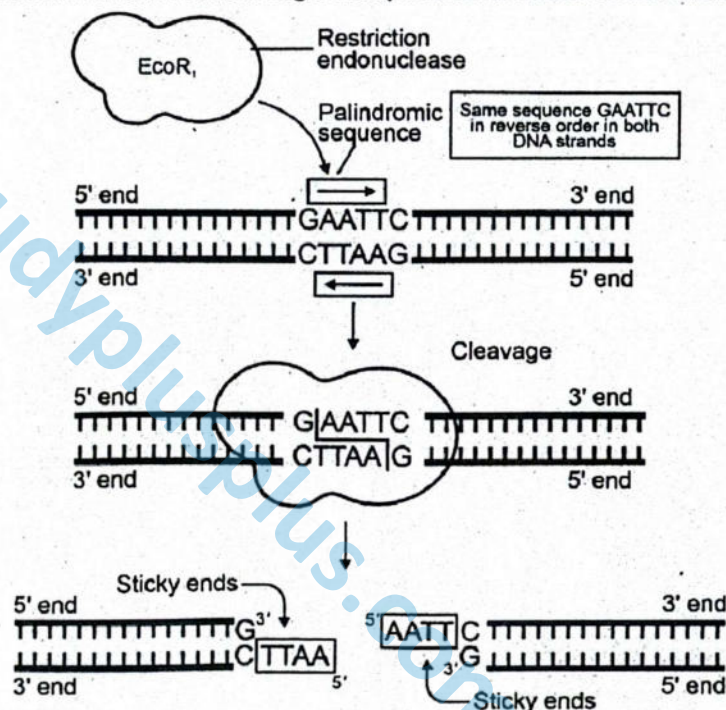


Figure: Recognition of palindromic sequence by restriction endonuclease *EcoRI* and production of sticky ends at 5' end in both complementary DNA strands

SLO [B-12-J-04]

Describe plasmid as vector prokaryotes and explain how recombinant plasmids can be formed.

PLASMIDS AS VECTORS IN PROKARYOTES

Plasmids are small, circular, double-stranded DNA molecules naturally found in many bacteria. Unlike chromosomal DNA, plasmids replicate independently and often carry genes that provide advantages to the host, such as antibiotic resistance or metabolic traits. In genetic engineering, plasmids are widely used as vectors—molecular carriers that transport foreign DNA into prokaryotic host cells for replication and expression.

► Key Features of Plasmid Vectors

Plasmids are preferred vectors because of their size, flexibility, and replication efficiency. Important features include:

1. Origin of Replication (*Ori*):

- Allows plasmids to replicate independently of the bacterial chromosome, producing multiple copies per cell.

2. **Selectable Markers:**
 - Genes such as antibiotic resistance markers (e.g., ampicillin or tetracycline) help identify bacteria that have successfully taken up the plasmid.
3. **Multiple Cloning Sites (MCS):**
 - Short sequences with unique restriction enzyme sites, allowing the precise insertion of foreign DNA.
4. **Promoter Regions:**
 - Enable transcription and expression of inserted genes within the host cell.
5. **Small Size:**
 - Facilitates easy manipulation in the laboratory.
6. **High Copy Number:**
 - Ensures **efficient amplification** of inserted genes, producing abundant recombinant DNA or protein.

Example: pBR322 is one of the earliest plasmids engineered for recombinant DNA studies:

- **Origin of replication:** Derived from pMB1.
- **Size:** 4,361 base pairs, compact and easy to manipulate.
- **Antibiotic resistance genes:** Ampicillin (Amp^R) and tetracycline (Tet^R).
- **Multiple restriction sites:** Enable insertion of foreign DNA into either Amp^R or Tet^R regions.

► **Formation of Recombinant Plasmids**

A **recombinant plasmid** is created when a **gene of interest** is inserted into a **plasmid vector**, forming a **hybrid DNA molecule** capable of replication and expression in a bacterial host. The process involves several steps:

1. **Isolation of the Gene of Interest**

- The target gene is obtained from genomic DNA or cDNA using **restriction enzymes**, which cut DNA at specific **palindromic sequences**, producing **sticky or blunt ends**.

2. **Preparation of the Plasmid Vector**

- The plasmid is cut with the **same restriction enzyme** as the gene of interest to produce complementary ends.
- This ensures that the gene can be inserted in the correct orientation.

3. **Ligation**

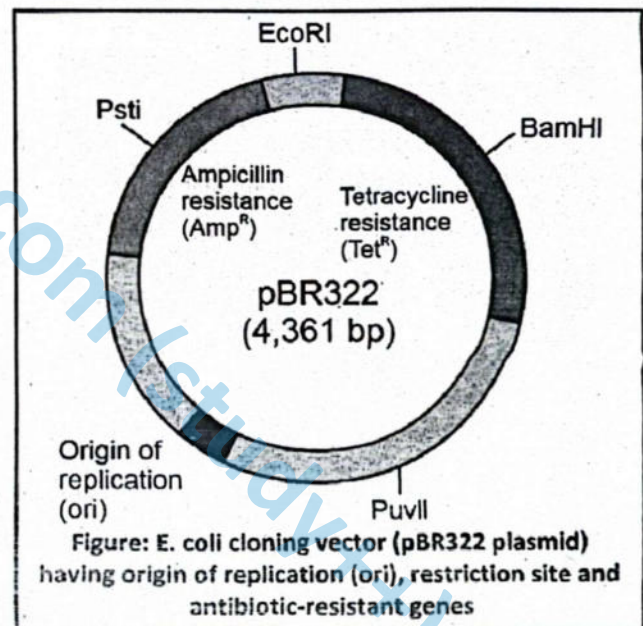
- The gene and plasmid are joined using DNA ligase, forming covalent bonds in the sugar-phosphate backbone.
- Sticky ends enhance the **efficiency and specificity of ligation**, while blunt ends allow flexible joining without overhangs.

4. **Transformation into Host Cells**

- The recombinant plasmid is introduced into **prokaryotic cells** (commonly *E. coli*) via **transformation techniques** such as heat shock or electroporation.
- Only bacteria that take up the recombinant plasmid survive if **selectable markers** are used (e.g., antibiotic resistance).

5. **Selection and Screening**

- Bacteria carrying recombinant plasmids are selected using **antibiotic-containing media**.
- Additional techniques, such as **blue-white screening**, can differentiate colonies containing recombinant plasmids from those with empty vectors.



Plasmids are versatile vectors in prokaryotes that allow scientists to replicate, express, and manipulate genes of interest. Through restriction enzyme cutting, ligation, and transformation, recombinant plasmids can be formed, making them essential tools in modern genetic engineering and biotechnology. They enable the production of medicines, industrial products, and genetically modified organisms, highlighting their critical role in science and industry.

SCIENCE TITBITS

Types of Plasmids as Vectors

1. **Cloning Vectors:** These are used for the cloning of DNA fragments. They contain essential elements like Ori, selectable markers, and cloning sites.
2. **Expression Vectors:** These are designed for the expression of a gene in a host organism. They include promoters, ribosome binding sites, and termination sequences.
3. **Shuttle Vectors:** Plasmids that can replicate in multiple host species, such as in both prokaryotic and eukaryotic cells, are known as shuttle vectors. These are useful for gene transfer between different biological systems.

SCIENCE TITBITS

Applications of Plasmids

- **Gene Cloning:** Plasmids are extensively used for cloning of genes of interest for further study and manipulation.
- **Protein Production:** Expression vectors are used for the production of proteins by expressing inserted genes.
- **Gene Therapy:** Plasmids are used to deliver therapeutic genes into cells of organisms with a genetic disease.
- **Vaccine Development:** Most of the DNA vaccines are Plasmid based and are effectively used to prevent diseases.

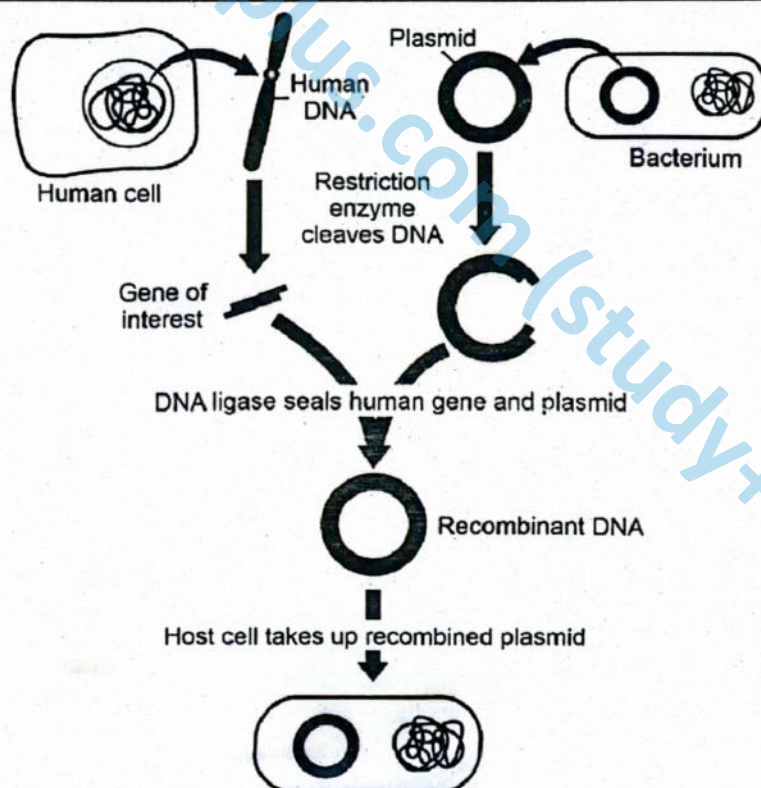


Figure: Steps for the formation of recombinant plasmid

PRODUCTION OF HUMAN INSULIN IN BACTERIA

The production of human insulin in bacteria is one of the most famous and important applications of genetic engineering. It allows diabetic patients to obtain insulin that is biologically identical to natural human insulin, avoiding the risks associated with animal-derived insulin. Bacteria such as *Escherichia coli* are used as bioreactors to produce insulin efficiently and on a large scale.

Steps in the Formation of Human Insulin in Bacteria

1. Identification and Isolation of the Insulin Gene

The insulin gene (INS), responsible for producing human insulin, is located on chromosome 11 at position 11p15.5. The gene spans approximately 1.5 kilobase pairs (Kbp), comprising three exons and two introns. Insulin is a peptide hormone composed of two chains—chain A (21 amino acids) and chain B (30 amino acids)—linked by disulfide bonds. It is naturally produced by the β -cells of the islets of Langerhans in the pancreas. To begin recombinant insulin production, the INS gene is identified and isolated from human DNA. However, since prokaryotes such as *Escherichia coli* cannot process eukaryotic introns, synthetic complementary DNA (cDNA) sequences or intron-free versions of the α and B chain genes are typically used.

2. Selection of suitable plasmid for insertion of insulin gene

Plasmids are circular DNA molecules used as vectors to carry foreign genes into bacterial cells. For insulin production, plasmids are chosen based on their compatibility with bacterial hosts and their features that support gene insertion and expression. Commonly used plasmids include:

- **pBR322:** A widely used plasmid containing antibiotic resistance genes for ampicillin and tetracycline. It has multiple restriction sites and a moderate copy number.
- **pUC18/pUC19:** Derivatives of pBR322, these plasmids are engineered for high-efficiency cloning. They include an ampicillin resistance gene and a multiple cloning site within the *lacZ* gene for blue-white screening.

These plasmids serve as carriers for the synthetic genes encoding the insulin chains. The A and B chain genes are inserted into plasmids separately and often adjacent to a bacterial gene such as *lacZ* (β -galactosidase), which facilitates the fusion and expression of insulin polypeptides.

3. Creation of Recombinant DNA

Bacteria e.g. *E. coli* are not so advanced cells so cannot directly produce eukaryotic proteins like insulin, so two separate synthetic genes are created for the α and B chains of insulin. These genes for both chains of insulin and separately inserted into two separate bacterial plasmid vectors i.e. *pBR322* plasmid. These genes are inserted into the plasmid by the side of *B* galactosidase gene. The plasmids are engineered with a promoter sequence that helps to initiate the auto- production of insulin chains when inside the bacterial cells.

SCIENCE TITBITS

Insulin was discovered by Edward S. Schafer in 1916. It is the major regulator of blood sugar in the body. Fredrick Sanger and his team in 1954 first time described the complete structure of insulin. Initially, B-cells synthesize pre-pro-insulin polypeptide, having 109 amino acids. Out of 109 amino acids, 23 amino acids act as signal molecules needed for entry of the pre-pro-insulin through cell membrane. When inside the cell, it become inactive pro-insulin with 86 amino acids, that finally converted to active insulin with 51 amino acids by some proteolytic enzymes. Active insulin consists of two polypeptide chains, chain a has 21 amino acids and chain B has 30 amino acids, both linked together by disulphide bonds.

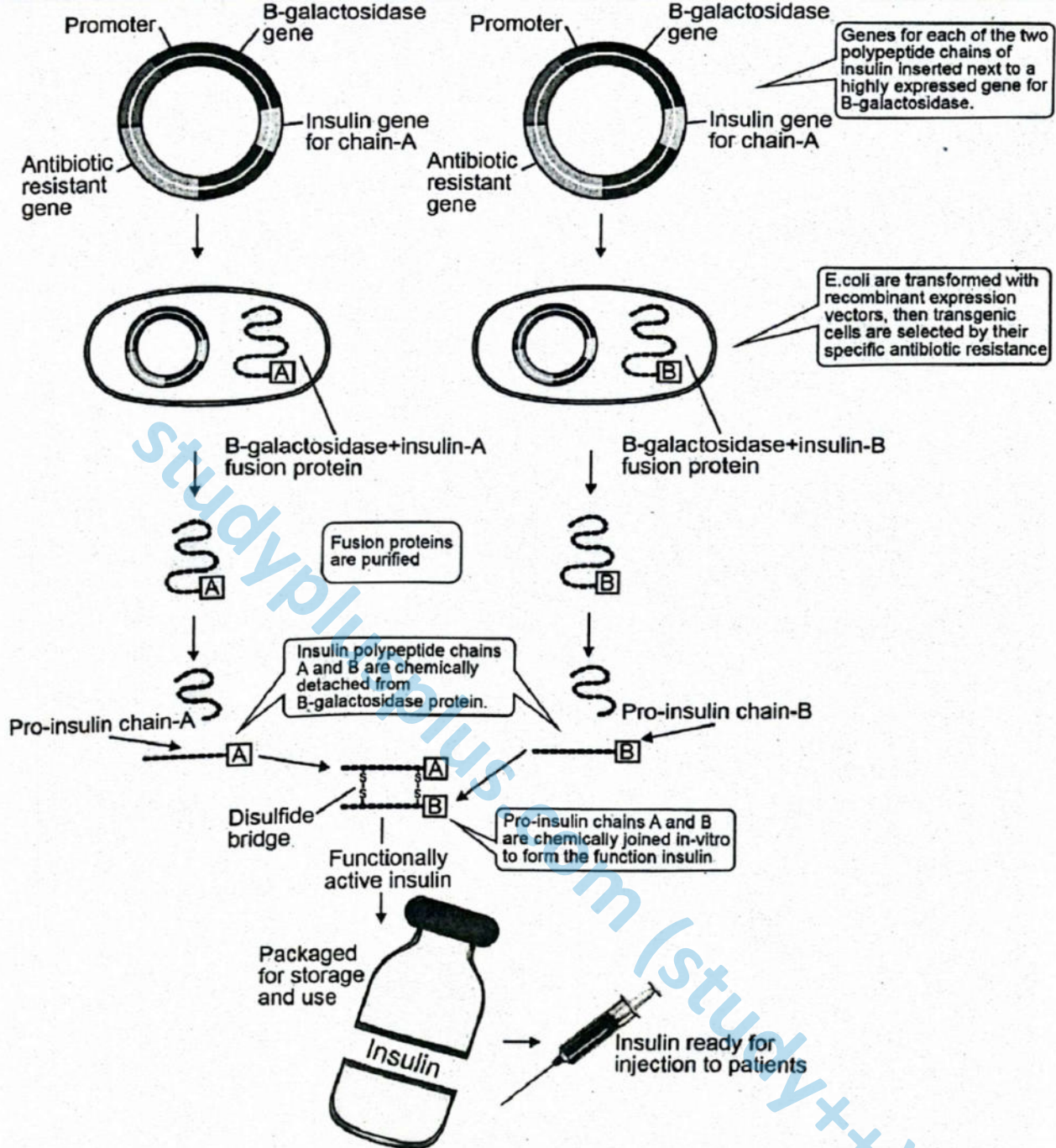


Figure: Steps for synthesis of human insulin in bacteria by recombinant DNA technology

4. Insertion of recombinant plasmids into host Cells

Once constructed, the recombinant plasmids carrying the genes for insulin A and B chains are introduced separately into competent *Escherichia coli* cells through a process known as **transformation**. This involves inducing the bacterial cells to take up plasmid DNA from their surroundings, often facilitated by calcium chloride treatment or electroporation. The transformed bacteria, now containing the recombinant plasmids, are referred to as **recombinant bacteria**.

5. Production of Insulin chains

The recombinant *E. coli* cells are cultured in large-scale **fermentation tanks** under sterile conditions, using nutrient-rich media and optimal environmental parameters (temperature, pH, aeration). As the bacteria multiply, the plasmid DNA replicates, resulting in the amplification (cloning) of the inserted insulin genes.

Each bacterial culture produces either the A or B chain of insulin as a **fusion protein** with β -galactosidase. This fusion approach enhances the stability of insulin peptides by protecting them from bacterial proteases during intracellular synthesis.

6. Extraction and Purification

For extraction of insulin, the bacterial cells are harvested and the pro-insulin chains combined with fusion proteins (β -galactosidase) are isolated. These pro-insulin chains A and B are separated from Fusion proteins, β -galactosidase by chemical treatment with cyanogen bromide. The removal of pro-insulin chains from fusion protein β -galactosidase is possible because, an extra codon is added before methionine codon at N-terminal of each gene for A and B-chain.

7. Folding and disulphide bridging of both insulin chains

Once a sufficient biomass is achieved, the bacterial cells are **harvested by centrifugation**, and the fusion proteins containing either the A or B insulin chains are isolated. The fusion proteins are engineered such that an additional codon encoding methionine is placed at the N-terminal of the insulin chain genes. This allows for the **chemical cleavage** of the fusion protein using **cyanogen bromide**, which selectively cleaves at methionine residues, thereby separating the insulin chains from the β -galactosidase component.

8. Purification and testing

The reconstituted insulin is **purified through a series of chromatographic techniques**, including ion-exchange and reverse-phase chromatography, to eliminate bacterial proteins, endotoxins, and other impurities. The final product undergoes **rigorous quality control tests**, including assessments of structural integrity, biological activity, sterility, safety, and the absence of adverse immunogenic reactions.

9. Packaging for storage, shipment and usage

Following quality validation, the purified recombinant human insulin is **formulated and packaged** into sterile vials, cartridges, or prefilled insulin pens. Packaging includes **labeling with instructions** for dosage, storage conditions (typically refrigeration), and method of administration. Recombinant human insulin produced in this manner is **structurally identical to natural human insulin**, offering superior purity, reduced risk of allergic reactions, and improved glycemic control compared to earlier animal-derived preparations.

SLO [B-12-J-02]

Explain polymerase chain reaction (PCR).

Polymerase Chain Reaction (PCR)

The **Polymerase Chain Reaction (PCR)** is a revolutionary molecular biology technique that allows scientists to **amplify specific segments of DNA** rapidly and precisely. Developed by **Kary Mullis in 1983**, PCR has become a cornerstone of **genetic research, diagnostics, and biotechnology**, enabling the study of genes from extremely small DNA samples.

Definition

PCR is a **method used to produce millions of copies of a specific DNA segment in vitro** (outside a living organism) using a thermostable **DNA polymerase, primers, nucleotides, and a thermal cycler**. This amplification allows even a single DNA molecule to be analyzed, cloned, or used in diagnostics.

► Principle of PCR

PCR relies on the natural process of DNA replication, but it is performed artificially in cycles. It involves:

1. Denaturation:

- Double-stranded DNA is heated to around **94–98°C**, causing the strands to separate into single strands.

2. Annealing:

- The temperature is lowered (**50–65°C**) to allow **short synthetic primers** to bind (anneal) to their **complementary sequences** on the target DNA.

3. Extension (Elongation):

- A **thermostable DNA polymerase** (commonly **Taq polymerase**) synthesizes new DNA strands by adding nucleotides complementary to the template.
- The extension is typically carried out at **72°C**, the optimal temperature for Taq polymerase activity.

This cycle is repeated **25–35 times**, resulting in **exponential amplification** of the target DNA segment.



Essential Components of a PCR Reaction Mixture

Polymerase Chain Reaction (PCR) requires a carefully prepared reaction mixture containing all the necessary molecular components for DNA amplification. The success of PCR depends on the correct composition and concentration of each element. These components include:

1. Taq DNA Polymerase

A thermostable DNA polymerase, **Taq polymerase** is the central enzyme used in PCR. It is isolated from the thermophilic bacterium *Thermus aquaticus*, which inhabits hot springs. This enzyme remains stable and catalytically active at high temperatures (up to 95 °C), making it ideal for the repeated thermal cycling required in PCR. Taq polymerase synthesizes new DNA strands by extending primers using deoxyribonucleotides.

2. Template DNA

The **template DNA** is the segment of DNA that contains the target sequence to be amplified. It provides the original sequence that the primers bind to and which is copied during the extension phase. The quantity and purity of template DNA affect the specificity and yield of the amplification.

3. Primers

Primers are short, single-stranded DNA oligonucleotides that are complementary to the sequences flanking the target DNA region. Two primers are required for each PCR reaction:

- Forward primer
- Reverse primer

They provide starting points for DNA synthesis by binding to opposite strands of the template DNA.

4. dNTPs (Deoxyribonucleoside Triphosphates)

These are the **building blocks** used by Taq polymerase to synthesize new DNA strands. The four types of dNTPs (dATP, dTTP, dGTP, and dCTP) must be present in balanced concentrations to ensure accurate and complete DNA replication.

5. Reaction Buffer

A **suitable buffer system** is essential to maintain the optimal pH and ionic strength required for the enzymatic activity of Taq polymerase. The buffer typically contains magnesium ions (Mg^{2+}), which act as cofactors for the enzyme. Some buffers may also include additives to enhance enzyme performance or DNA stability.

6. Thermocycler (PCR Machine)

The **thermocycler**, also known as a PCR machine, is an automated instrument programmed to precisely control the temperature changes required during the PCR cycles—denaturation, annealing, and extension. It ensures rapid and accurate temperature transitions, critical for the efficiency and specificity of the reaction.

Mechanism of PCR Reaction

PCR involves a series of thermal cycling steps that enable the selective and exponential amplification of a DNA fragment. Each cycle typically includes three main steps:

1. Denaturation
2. Primer annealing
3. Extension (polymerization)

These steps are carried out using a programmable device called a **thermal cycler (thermocycler)**, which precisely regulates temperature and timing.

1. Denaturation

a. Initial Denaturation

- **Purpose:** To fully denature the DNA template by breaking hydrogen bonds between the two strands.
- **Conditions:** 94–95°C for 4–5 minutes.
- **Rationale:** Necessary to disrupt stable secondary structures, especially in GC-rich regions, and to ensure that the entire template becomes single-stranded before cyclic amplification begins.

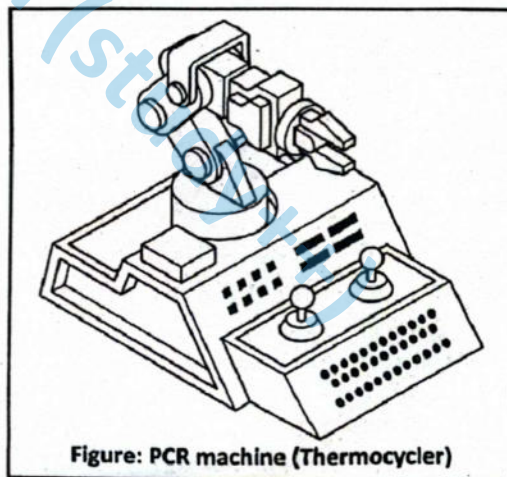


Figure: PCR machine (Thermocycler)

b. Cyclic Denaturation

- **Purpose:** To separate DNA strands before each amplification cycle.
- **Conditions:** 94 °C for ~1 minute.
- **Outcome:** Double-stranded DNA (dsDNA) becomes single-stranded DNA (ssDNA), providing a template for primer binding and extension in subsequent steps.

2. Primer Annealing

- **Purpose:** To allow short, sequence-specific primers (forward and reverse) to bind to their complementary regions on the single-stranded DNA templates.
- **Conditions:** 55–65 °C for ~2 minutes (temperature depends on primer length and GC content).
- **Outcome:** Primers hybridize to their target sequences, flanking the DNA region to be amplified.

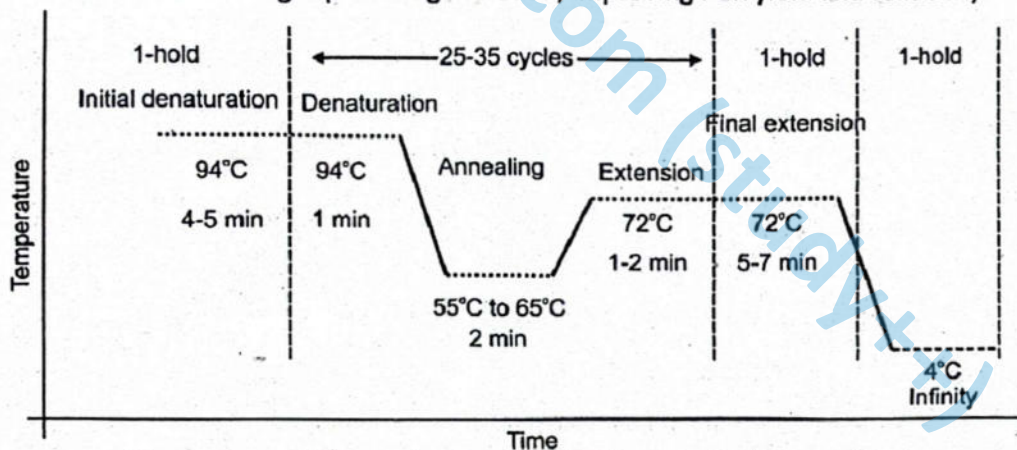
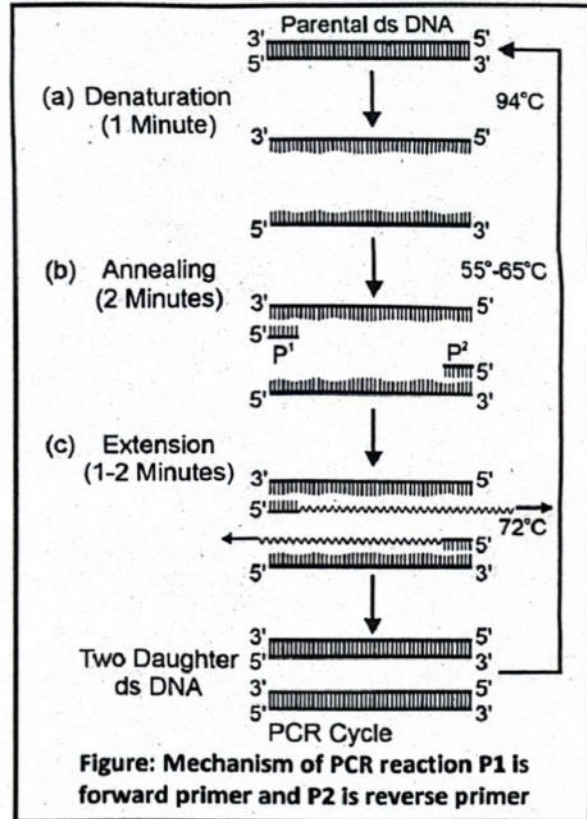
3. Extension or Polymerization

Cyclic Extension

- **Purpose:** To synthesize new DNA strands using the primers as starting points.
- **Conditions:** 72 °C for ~1 minute per cycle.
- **Enzyme:** **Taq DNA polymerase**, a thermostable enzyme isolated from *Thermus aquaticus*, catalyzes the addition of deoxyribonucleotides (dNTPs) to the 3' end of each primer.
- **Outcome:** Complementary strands are synthesized, forming new double-stranded DNA.

Final Extension

- **Purpose:** To complete any partially synthesized DNA strands.
- **Conditions:** 72 °C for 5–7 minutes (performed once, after all cycles).
- **Rationale:** Ensures full-length product generation, improving PCR yield and reliability.



► Cycle Repetition and Amplification

At the end of the **first cycle**, one double-stranded DNA (dsDNA) molecule becomes two. In the **second cycle**, each of the two DNA molecules is denatured again and serves as a template, producing four DNA copies, and so on. This results in **exponential amplification**, with the DNA quantity doubling after every cycle.

The process is typically repeated for **25–35 cycles**, depending on the desired amount of amplified DNA.

► Storage of PCR Products

After completion of the final extension, the reaction is held at **4°C** to preserve the PCR products. This temperature prevents degradation of amplified DNA until it is retrieved and stored in a refrigerator or used for downstream applications such as gel electrophoresis, cloning, or sequencing.

Step	Purpose	Typical Temperature	Duration
Initial Denaturation	Denature template DNA completely	94–95 °C	4–5 minutes
Cyclic Denaturation	Separate dsDNA into ssDNA	94 °C	~1 minute
Primer Annealing	Allow primers to bind to complementary regions	55–65 °C	~2 minutes
Extension	Synthesize new DNA strands with Taq polymerase	72 °C	~1 minute per cycle
Final Extension	Complete any unfinished DNA synthesis	72 °C	5–7 minutes
Hold/Storage	Preserve PCR products	4 °C	Indefinite (until use)

SCIENCE TITBITS

Q. Why heat is used in PCR technique to denature the target DNA instead of using DNA helicase and DNA gyrase enzymes?

Ans. Heat is used in PCR to break the hydrogen bonds between DNA strands because enzymes like helicase and gyrase are not stable at high temperatures and cannot function efficiently *in vitro*. Heating to ~94°C rapidly separates the DNA strands without the need for enzymatic unwinding.

Q. Why human DNA polymerase cannot be used in PCR technique?

Ans. Human DNA polymerase is not thermostable and becomes denatured at the high temperatures required during the denaturation step of PCR. In contrast, Taq polymerase from *Thermus aquaticus* remains active at high temperatures, making it suitable for thermal cycling in PCR.

Q. Why already synthesized primers are used in PCR technique instead of using primase enzyme?

Ans. Primase is not used in PCR because it functions only *in vivo* and is not stable during high-temperature cycles. Instead, synthetic DNA primers are added to the reaction. These primers are designed to specifically bind to the target sequence and remain intact throughout the thermal cycling process.

SLO [B-12-J-05]

Define genetically modified organism.

GENETICALLY MODIFIED ORGANISMS (GMOS)

A **Genetically Modified Organism (GMO)** is an organism whose **genetic material (DNA)** has been **deliberately altered using modern biotechnology techniques**. This modification is done to **introduce, remove, or edit specific genes**, resulting in traits that may not occur naturally in the organism.

Definition

"An organism whose genome has been altered by the insertion, deletion, or modification of specific genes using recombinant DNA technology, gene editing, or other molecular techniques."

► Purpose of Creating GMOs

Genetic modification allows scientists to **improve organisms** for human benefit by:

1. Enhancing Agricultural Traits:

- Crops that are pest-resistant, drought-tolerant, or nutritionally enriched.
- **Examples:** Bt cotton (insect-resistant), Golden Rice (vitamin A enriched).

2. Medical Applications:

- Bacteria engineered to produce human insulin, growth hormones, and vaccines.

3. Industrial and Environmental Uses:

- Microbes modified for biofuel production, biodegradation of pollutants, or industrial enzyme synthesis.



► Key Characteristics of GMOs

Genetically Modified Organisms (GMOs) are unique because of several defining characteristics that distinguish them from naturally occurring organisms. These features make GMOs highly valuable in agriculture, medicine, and biotechnology.

1. Precision:

- Genetic modifications in GMOs are **highly specific**, targeting only the **desired genes**.
- Modern techniques such as **recombinant DNA technology, CRISPR-Cas9, and gene cloning** allow scientists to **add, remove, or edit particular genes** without affecting other parts of the genome.
- **Example:** In Bt cotton, the gene for insect resistance is precisely inserted without altering other plant traits like growth or fiber quality.

2. Novel Traits:

- GMOs often exhibit traits that **do not naturally occur** in the species.
- These **new characteristics** can provide **enhanced nutrition, resistance to pests or diseases, faster growth, or environmental tolerance**.
- **Example:** Golden Rice is genetically modified to produce **beta-carotene (vitamin A precursor)**, a trait not naturally found in traditional rice varieties.

3. Reproducibility:

- Once a genetic modification is successfully incorporated, the **trait can be stably inherited** by the next generation, provided the modification is integrated into the organism's germline.
- This allows GMOs to **maintain their enhanced characteristics across generations**, ensuring long-term benefits.
- **Example:** Bt corn produces the insecticidal protein in every generation, **protecting crops consistently** without repeated intervention.

4. Versatility and Adaptability:

- GMOs can be designed for specific environments or purposes, making them highly adaptable for **different climates, industrial needs, or medical applications**.
- **Example:** Engineered bacteria can survive in industrial reactors to produce **biofuels or pharmaceuticals efficiently**.

5. Controllability:

- Scientists can **control when and where the introduced genes are expressed** using regulatory sequences such as promoters and enhancers.
- This ensures that the GMO expresses the desired trait only under certain conditions or in specific tissues.
- **Example:** Pest-resistant genes may be expressed only in leaves to target herbivorous insects without affecting other plant parts.

The key characteristics of GMOs—**precision, novel traits, reproducibility, versatility, and controllability**—make them powerful tools in modern biotechnology. These features enable the creation of organisms with **enhanced or entirely new properties**, offering solutions for **food security, medical therapies, industrial processes, and environmental sustainability**.

Examples of GMOs

Type	Example	Purpose
Plants	Bt cotton, Golden Rice	Pest resistance, enhanced nutrition
Animals	AquAdvantage salmon	Faster growth
Microorganisms	E. coli producing human insulin	Medicine production



Both strains of *Pseudomonas syringae* occur naturally, but recombinant DNA technology has allowed for the synthetic removal or alteration of specific genes, enabling the creation of the ice-minus strain. Modifying *P. syringae* may have unexpected consequences for climate. A study has shown that its ice nucleating proteins may play an important part in causing ice crystals to form in clouds. If humans increase the frequency of bacteria lacking these proteins then it could potentially affect rainfall.

The first genes available for genetic engineering of crop plants for pest resistance were known as Bt genes from a bacterium *Bacillus thuringiensis*. These are specific to particular group of insect pests and are not harmful to other useful insects like butter flies and silk worms. Transgenic crops with Bt genes (e.g. cotton, rice, maize, potato, tomato, brinjal, cauliflower, cabbage, etc.) have been developed. This has proved to be an effective way of controlling the insect pests and has reduced the pesticide use.

Concerns about the Genetically Modified Organisms (GMOS)

The main areas of consideration for safety aspects in biotechnology are the following:

- How to dispose-off spent microbial biomass and purify the effluents from biotechnological processes?
- The toxicity of the allergy associated with microbial production.
- How to deal with the increase in the number of antibiotic resistant pathogenic microorganisms?
- How to evaluate the pathogenicity of the genetically engineered microorganisms to infect humans, plants and animals?
- How to prevent contamination, infection or mutation of the processed strains?

1. Transgenic Bacteria

Due to their simple genetic makeup and rapid growth, bacteria are highly amenable to genetic transformation. The first successful example was reported in 1978 when the human insulin gene was inserted into *Escherichia coli*, enabling the bacterial synthesis of recombinant human insulin. Since then, transgenic bacteria have been widely employed not only in the production of therapeutic human proteins but also in diverse applications such as enhancing plant growth, bioremediation of environmental pollutants, and bioleaching of metals from low-grade ores.

2. Transgenic Plants

The first field trials of genetically modified (GM) plants took place in 1986 in France and the United States, using herbicide-resistant tobacco plants. The primary goal of developing transgenic plants is to introduce beneficial traits that do not naturally exist within the species. These traits include resistance to pests, diseases, herbicides, or environmental stressors, as well as the ability to produce enhanced nutrients or pharmaceutical compounds.

A landmark advancement in crop protection was the introduction of Bt genes from the soil bacterium *Bacillus thuringiensis*. These genes code for proteins that are toxic to specific insect pests but are harmless to beneficial insects such as butterflies and silkworms. Bt genes have been incorporated into various crops, including cotton, maize, rice, potato, tomato, brinjal, cauliflower, and cabbage. These genetically modified crops have significantly reduced the need for chemical pesticides and improved pest resistance.

3. Transgenic Animals

Transgenic animals are those into which foreign genes have been deliberately introduced using recombinant DNA technology. Genetic engineering in animals aims to enhance desirable traits in livestock, such as growth rate, milk production, or disease resistance. Moreover, certain transgenic animals have been engineered to produce pharmaceutical proteins in their milk, blood, or eggs. These animals, known as "transpharmer" animals, serve as bioreactors for the cost-effective production of biologically active human drugs.

SCIENCE, TECHNOLOGY AND SOCIETY CONNECTIONS

Global Transgenic Crops

Since 1996, the cultivation of transgenic crops has expanded rapidly. In 2005, transgenic crops were planted globally on 90 million hectares, which increased to 190.4 million hectares by 2019. Major GM crops include herbicide tolerance and insect resistance soybean, maize, cotton, and canola. Leading producers are the United States, Brazil, Argentina and Canada. Transgenic Crops in Pakistan commercially adopted Bt cotton in 2010, which is now grown over 2.3 million hectares (95% of the cotton area). New triple gene cotton varieties, resistant to pests and herbicides, have been introduced in 2023. GM maize production increased sharply, from 1 million metric tons in 2013 to 10.5 million metric tons in 2023. Recently, GM varieties of sugarcane has also been approved for cultivation. (Discourse: 2024, PIDE, Islamabad)



Most of the countries of the world are signatories to the Biological Weapons Conventions of 1972. As a signatory, it is a voluntary pledge by a nation "never to produce microbial or other biological agents or toxins, whatever may be their method of production, for use in wars. However, many people have expressed their concerns about the possible use of genetic manipulations for military purposes in the near future.

SCIENCE, TECHNOLOGY AND SOCIETY CONNECTIONS

- **Justify the need of genetic counselling.**

Genetic counselling is a service that provides information and advice about genetic conditions. Counselling is conducted by healthcare professionals who have been specially trained in the science of human genetics (a genetic counsellor or a clinical geneticist). The counsellor will discuss the risks, benefits and limitations of genetic testing with you. They will also explain how the information found as a result of genetic testing could have implications for both you and your family.

- **Investigate careers that require an understanding of biotechnology and genetic engineering.**

Many careers require an understanding of biotechnology and genetic engineering. Such as: Healthcare professionals, Teachers of biological sciences, biomedical engineers, crime lab analyst, crime scene investigator, environmental impact analyst, forensic scientist, genetic engineer, molecular biologist etc.

- **Describe briefly the accomplishments of the renowned genetic engineers working in private and public sector institutions in her or his province.**

It is advised to the administration that they should arrange study tour of student to public and private sector institutions of genetic engineering so that students can meet with renowned genetic engineers and can directly ask about their accomplishments.

- **Suggest measure she/he would take to solve related problems by using knowledge gained in this chapter.**

Students should evaluate themselves to know what they have learnt in this chapter and should apply this knowledge to solve related problems under the guidance of their class teacher.

- **Describe the role of genetic screening.**

Genetic screening includes all those diagnostics tests which are used to determine whether a person or a new born baby is at risk of genetic diseases or not. Generally there are two types of genetic screening, screening of children and adults, and screening of unborn children. Genetic screening of children and adults has two purposes: first it can confirm whether the person has a mutated gene of certain disease or characteristics. The second purpose is to test adults to see if their children will be at risk of certain disease. Knowing that one or both parents carries a dominant allele for a genetic disease might affect the decision of parents about having birth of affected children. Now a days, genetic screening is used for approval of marriage licenses in some countries, like Denmark.

SLOs Based (MCQs)

GENETIC ENGINEERING AND PCR

- | | |
|---|---|
| <p>1. Which of the following is used as a vector in genetic engineering?</p> <p>A. Ribosome B. Plasmid
C. Nucleosome D. Lysosome</p> <p>2. The term "recombinant DNA" refers to:</p> <p>A. DNA extracted from recombinant proteins
B. DNA that mutates naturally
C. DNA formed by combining genetic material from two different sources
D. DNA synthesized by ribosomes</p> | <p>3. Which enzyme is used to cut DNA at specific sequences in genetic engineering?</p> <p>A. DNA ligase
B. Restriction endonuclease
C. DNA polymerase D. Helicase</p> <p>4. Which of the following is not a feature of a plasmid used in genetic engineering?</p> <p>A. Origin of replication B. Restriction sites
C. Ribosomal genes
D. Antibiotic resistance genes</p> |
|---|---|

5. In insulin gene cloning, which organism is commonly used as a host cell?
A. *Saccharomyces cerevisiae*
B. *E. coli*
C. *Bacillus subtilis*
D. *Rhizobium*
6. Which of the following enzymes joins DNA fragments?
A. Reverse transcriptase
B. RNA polymerase
C. DNA ligase
D. DNA helicase
7. A gene inserted next to the *lacZ* gene in a plasmid allows:
A. Elimination of unwanted proteins
B. Expression as a fusion protein
C. Replication of the host cell
D. Cell wall synthesis
8. The codon inserted before the methionine codon in insulin gene expression is to:
A. Start protein synthesis
B. Improve ribosome binding
C. Facilitate cleavage by cyanogen bromide
D. Stop protein synthesis
9. Recombinant human insulin differs from animal insulin by:
A. Having a different amino acid sequence
B. Being inactive
C. Being less pure
D. Being identical to natural human insulin
10. The restriction enzyme *EcoRI* recognizes which sequence?
A. AGCT
B. GAATTC
C. AAGGTT
D. GCGC
11. Which enzyme is used in PCR?
A. Helicase
B. DNA polymerase I
C. Taq polymerase
D. RNA polymerase
12. PCR is primarily used to:
A. Join DNA fragments
B. Clone cells
C. Amplify DNA
D. Transcribe RNA
13. What is the typical temperature for primer annealing in PCR?
A. 25°C
B. 94°C
C. 72°C
D. 55–65°C
14. The initial denaturation step in PCR is carried out:
A. In every cycle
B. Only once at the beginning
C. At the end of the process
D. In alternate cycles
15. The final extension step ensures:
A. DNA degradation
B. Primer removal
C. Complete synthesis of DNA strands
D. Transformation
16. Cyanogen bromide is used in insulin production to:
A. Break disulfide bonds
B. Join DNA fragments
C. Cleave fusion proteins at methionine residues
D. Denature DNA
17. Which of the following is an essential component of a PCR reaction?
A. RNA polymerase
B. Ligase
C. dNTPs
D. Ribosomes
18. What is the optimal temperature for Taq polymerase activity?
A. 55°C
B. 94°C
C. 72°C
D. 37°C
19. Which feature of Taq polymerase makes it suitable for PCR?
A. It synthesizes RNA
B. It is stable at high temperatures
C. It breaks down proteins
D. It removes introns
20. What happens to DNA during the denaturation step of PCR?
A. It is transcribed into RNA
B. It is broken down into nucleotides
C. It becomes double-stranded
D. It separates into single strands
21. PCR differs from natural DNA replication in that it:
A. Requires a host cell
B. Does not need primers
C. Is performed *in vitro*
D. Occurs only during cell division
22. The exponential nature of PCR means:
A. Each cycle produces linear DNA increase
B. DNA amount remains constant
C. DNA quantity doubles with each cycle
D. Only one strand is amplified
23. What is the purpose of primers in PCR?
A. To synthesize new DNA
B. To break DNA strands
C. To bind and initiate DNA synthesis
D. To degrade unwanted DNA
24. Which of the following occurs at 94°C in PCR?
A. Primer binding
B. DNA strand synthesis
C. DNA denaturation
D. Protein folding
25. After PCR, DNA products are stored at:
A. 94°C
B. Room temperature
C. 4°C
D. 100°C
26. The number of DNA molecules after 10 PCR cycles from 1 copy is:
A. 100
B. 512
C. 1024
D. 2048



27. In PCR, the final product is:
A. RNA copies B. Protein fragments
C. Amplified DNA D. Fusion proteins
28. The thermocycler is used in PCR to:
A. Separate proteins
B. Maintain bacterial cultures
C. Regulate temperature for DNA amplification
D. Clone cells

29. The first step of PCR cycle is:
A. Annealing B. Denaturation
C. Extension D. Replication
30. PCR is especially useful in:
A. Protein synthesis
B. DNA fingerprinting and diagnostics
C. Lipid metabolism
D. Photosynthesis studies

Answer Key

1. B	2. C	3. B	4. C	5. B	6. C	7. B	8. C	9. D	10. B	11. C	12. C	13. D	14. B	15. C
16. C	17. C	18. C	19. B	20. D	21. C	22. C	23. C	24. C	25. C	26. C	27. C	28. C	29. B	30. B

Short Questions

Q.1 What is recombinant DNA technology?

Ans. Recombinant DNA technology is the process of combining DNA from two different sources to create a novel DNA molecule. This technology allows specific genes to be inserted into vectors such as plasmids, which can then be introduced into host organisms like bacteria or yeast. Once inside the host, the gene can be expressed, enabling the production of desired proteins, enzymes, or hormones. A key application is the production of human insulin, where synthetic insulin genes are expressed in *E. coli* to produce large quantities of the hormone for diabetic patients. This technology also underpins vaccine development, gene therapy, and agricultural biotechnology, making it a cornerstone of modern molecular biology.

MARKING SCHEME (3 Marks)

- Definition – 2 marks, uses – 1 mark

Q.2 What are the basic steps of genetic engineering?

Ans. The main steps of genetic engineering begin with the identification and isolation of a gene of interest from a donor organism. Next, the gene is inserted into a suitable vector, such as a plasmid, using molecular tools like restriction enzymes and ligases. The recombinant vector is then introduced into a host organism through transformation, allowing the host to replicate and express the gene. The expressed protein or product is subsequently extracted, purified, and tested for activity. These steps enable scientists to manipulate genetic material for applications such as producing therapeutic proteins, enhancing crop traits, or studying gene function, forming the foundation of biotechnology and molecular medicine.

MARKING SCHEME (3 Marks)

- Names of all steps – 3 marks

Q.3 Why is *E. coli* used in genetic engineering?

Ans. *E. coli* is widely used as a host in genetic engineering due to its well-characterized genetics, rapid growth rate, and ability to efficiently replicate plasmid DNA. Its cells are easy to transform with foreign DNA, and they can express inserted genes at high levels. Additionally, *E. coli* can be cultured inexpensively in large quantities, making it ideal for industrial-scale protein production. Its safety profile and extensive research background provide a reliable platform for gene cloning, protein expression, and recombinant DNA studies. These properties make *E. coli* one of the most commonly used organisms in molecular biology, biotechnology, and pharmaceutical manufacturing.

MARKING SCHEME (3 Marks)

- Properties of *E. coli* useful as a vector – 3 marks

Q.4 How is human insulin produced using recombinant DNA?

Ans. Human insulin production using recombinant DNA involves inserting synthetic genes for insulin chains A and B into separate plasmids, which are introduced into *E. coli* cells. The bacteria express the chains as fusion proteins, which are then extracted and purified. The fusion proteins are cleaved to release the individual A and B chains, which are chemically combined to form active insulin identical to human insulin. This method allows mass production of insulin that is safe, consistent, and cost-effective compared to earlier methods using animal pancreas extracts. Recombinant insulin has transformed diabetes treatment and illustrates the power of genetic engineering in producing therapeutic proteins.

MARKING SCHEME (3 Marks)

- Names of steps involved – 3 marks



Q.5 What is the role of restriction enzymes in genetic engineering?

Ans. Restriction enzymes, also called restriction endonucleases, are proteins that cut DNA at specific nucleotide sequences. They act like molecular scissors, enabling precise cutting of DNA fragments. In genetic engineering, these enzymes allow scientists to remove or insert genes into vectors, forming recombinant DNA molecules. By generating compatible ends on DNA fragments and vectors, restriction enzymes facilitate the ligation and assembly of new genetic constructs. They are essential tools for cloning, gene expression studies, and creating genetically modified organisms. Without restriction enzymes, targeted manipulation of DNA for research, medicine, or biotechnology would be nearly impossible.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, role – 2 marks

Q.6 Define Polymerase Chain Reaction (PCR).

Ans. Polymerase Chain Reaction (PCR) is a laboratory technique used to amplify a specific DNA sequence, generating millions of copies from a single or few DNA molecules. PCR involves repeated cycles of denaturation, where the double-stranded DNA is separated; annealing, where short primers bind to complementary sequences on the target DNA; and extension, where a DNA polymerase synthesizes new strands starting from the primers. This cyclical process allows researchers to obtain large amounts of DNA from minimal starting material, facilitating applications such as genetic testing, forensic analysis, cloning, and pathogen detection. PCR is fundamental in molecular biology due to its speed, specificity, and sensitivity.

MARKING SCHEME (3 Marks)

- Definition – 2 marks, names of steps – 1 mark

Q.7 Why is Taq polymerase used in PCR?

Ans. Taq polymerase is a DNA polymerase derived from the thermophilic bacterium *Thermus aquaticus*. It is thermostable, meaning it remains active at the high temperatures required for DNA denaturation during PCR. This stability allows the enzyme to repeatedly synthesize new DNA strands during successive cycles of amplification without being denatured itself. Taq polymerase efficiently extends primers to produce complementary DNA strands, enabling the exponential amplification of the target sequence. Its robustness and efficiency make it the enzyme of choice for PCR, allowing precise, rapid, and reliable DNA replication for applications such as diagnostics, cloning, and forensic analysis.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, importance – 2 marks

Q.8 What is the function of primers in PCR?

Ans. Primers are short, single-stranded DNA sequences that are complementary to regions flanking the target DNA segment. They serve as starting points for DNA synthesis by providing free 3'-OH ends for Taq polymerase to attach nucleotides. During the annealing step of PCR, primers specifically bind to their complementary sequences on the template DNA, ensuring that only the desired region is amplified. Without primers, DNA polymerase cannot initiate synthesis. Primers thus provide specificity to PCR, define the boundaries of the amplified sequence, and determine the efficiency and accuracy of the reaction, making them essential components for successful DNA amplification.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, function – 2 marks

Q.9 How does PCR achieve exponential amplification of DNA?

Ans. PCR achieves exponential amplification because each cycle doubles the number of DNA molecules. In the first cycle, the original DNA template produces two copies. In subsequent cycles, both original and newly synthesized strands serve as templates for further amplification. This doubling process continues through multiple cycles, resulting in millions of copies of the target DNA from a minimal starting amount. The combination of repeated denaturation, primer annealing, and extension ensures that the reaction is both rapid and highly specific. Exponential amplification enables sensitive detection of genetic material in applications such as disease diagnosis, forensic analysis, and molecular cloning.

MARKING SCHEME (3 Marks)

- Definition of amplification – 1 mark, reason for exponential amplification – 2 marks

Q.10 What is the purpose of the final extension step in PCR?

Ans. The final extension step in PCR allows Taq polymerase to complete the synthesis of any partially extended DNA strands remaining from previous cycles. During this step, the reaction is held at an optimal temperature, enabling the polymerase to add nucleotides to the 3' ends

MARKING SCHEME (3 Marks)

- Definition of extension – 1 mark, purpose – 2 marks



of all newly synthesized DNA strands. This ensures that the PCR products are fully double-stranded, complete, and uniform in length. The final extension maximizes yield and accuracy of the amplified DNA, which is crucial for downstream applications such as cloning, sequencing, or diagnostic assays, where incomplete DNA products could compromise results.

SLO [B-12-J-07]

Describe how vertical food farms (soil free) work.

WORKING OF VERTICAL (SOIL-FREE) FOOD FARMS

Vertical food farms, also called **vertical farming**, are an innovative method of growing crops **without traditional soil**, using controlled **soilless systems** in stacked layers. This approach allows **year-round cultivation**, maximizes space efficiency, and reduces reliance on conventional agriculture, making it ideal for **urban environments**.

Vertical farming is the practice of growing plants in vertically stacked layers using soilless techniques such as hydroponics, aeroponics, or aquaponics, under controlled environmental conditions including light, temperature, and nutrients.

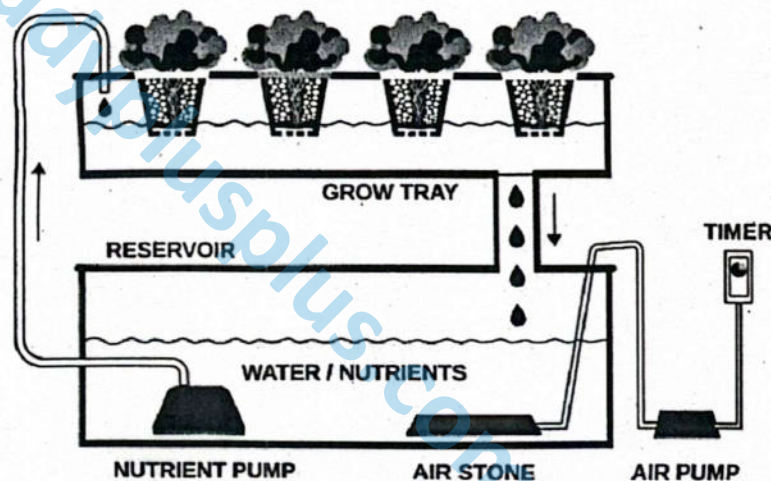


Figure: Working of vertical (Soil-free) food farms

How Soil-Free Vertical Farms Work

Soilless Growth Systems

Vertical farms eliminate soil by using alternative growth methods:

1. Hydroponics

In hydroponics, plants are grown with their roots immersed in a nutrient-rich water solution instead of soil. The system circulates water through channels, delivering nutrients directly to the plant roots and recycling it to conserve resources—using up to 90% less water than traditional farming methods.

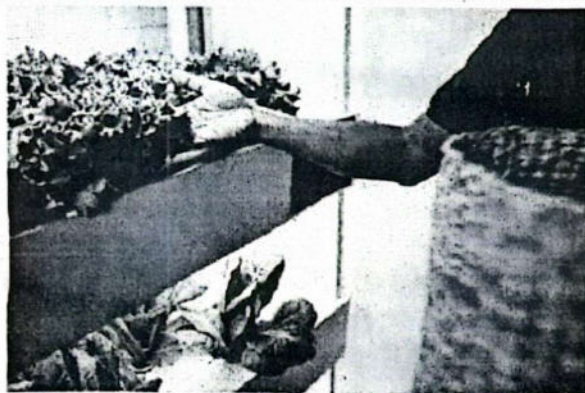


Figure: Harvesting of fruits from vertical (Soil-free) food farms

2. Aeroponics

Aeroponics involves suspending plant roots in air, where they are periodically misted with nutrient-rich water. This method enhances oxygen availability to roots and supports efficient absorption of nutrients and water. Aeroponics offers greater water efficiency, improved oxygenation, and higher crop yields compared to hydroponics.

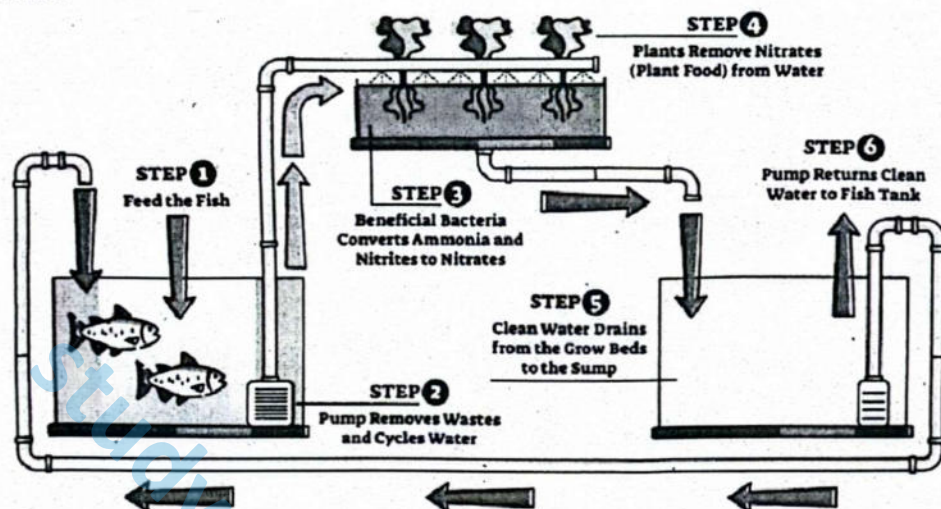


Figure: Schematic diagram of Aquaponics: Vertical (Soil-free) food farming with fish farming

3. Aquaponics

Aquaponics combines hydroponics with aquaculture (fish farming). In this system, water from fish tanks is cycled through plant beds. Fish waste serves as a natural fertilizer for plants, while the plants purify the water for reuse in the fish tanks. This creates a sustainable, closed-loop ecosystem with minimal need for synthetic inputs.



Figure: Well established large scale vertical (Soil-free) food farm

Vertical farming is supported by **Controlled Environment Agriculture (CEA)**, which utilizes technology to maintain optimal growing conditions year-round. CEA systems include:

- **LED grow lights** with programmable timers to simulate appropriate light spectra for photosynthesis.
- **Climate control systems** to regulate temperature and humidity.
- **Sensor networks** to monitor and adjust nutrient delivery, light exposure, and overall plant health.

This method enables consistent, rapid plant growth and reliable harvests throughout the year while minimizing land and water usage. Vertical farming is considered more sustainable than traditional agriculture, as it significantly reduces water consumption, greenhouse gas emissions, and land use, while yielding more crops per unit area.

Modern vertical farms are also integrated with **automated sensor systems and artificial intelligence (AI)**, which optimize production by monitoring plant growth, adjusting environmental conditions, and managing nutrient delivery with precision.

COMPARISON OF VERTICAL FOOD FARMS AND TRADITIONAL AGRICULTURE IN PAKISTAN

Agriculture is a cornerstone of Pakistan's economy, providing employment to a large proportion of the rural population and serving as a major contributor to the country's GDP. It supplies raw materials for key industries such as textiles, sugar, food processing, and dairy.

► Main Features of Traditional Agriculture in Pakistan

- (i) **Agricultural Land:** Pakistan possesses 47% arable land, significantly higher than the global average of 38%. About 82% of cultivated land is irrigated through an extensive canal system, while 18% is rain-fed or arid.
- (ii) **Major Crops:** Staple crops include wheat, rice, sugarcane, and potatoes, while cotton and maize are prominent cash crops. The cropping calendar consists of two main seasons: *Kharif* and *Rabi*.
- (iii) **Livestock Sector:** Livestock complements crop production and is a major economic sector. Pakistan is the fifth-largest producer of dairy products and the fourth-largest exporter of leather goods globally.
- (iv) **Organic Farming:** Organic practices are observed through the use of recycled crop residues and animal manure, minimal use of synthetic pesticides, and implementation of crop rotation to reduce environmental impact.

Table 10.1: Comparison Between Vertical Farming and Traditional Agricultural Practices

Aspect	Vertical Food Farming	Traditional Agricultural Practices
Space Utilization	Utilizes vertical space by stacking crops in multiple layers, allowing cultivation in urban areas with limited land availability. Particularly suitable for densely populated regions.	Requires extensive horizontal land, making it more feasible in rural areas. Staple crops such as wheat, rice, and sugarcane demand large tracts of land, as commonly seen in the agricultural zones of Punjab and Sindh.
Water Usage	Uses 90–95% less water than conventional farming. Systems such as hydroponics and aeroponics recycle or mist water directly to roots, significantly minimizing water loss. This approach is especially relevant in arid regions like Thar and Balochistan.	Requires large volumes of water, particularly for crops like rice and sugarcane. Traditional irrigation methods often result in wastage due to seepage and flood-based overflow, contributing to water scarcity and environmental stress.
Yield and Productivity	Provides higher yields per square foot due to controlled environmental conditions (light, temperature, humidity, and nutrient delivery). Enables year-round production of crops such as leafy vegetables, herbs, and strawberries.	Productivity is seasonal and dependent on environmental factors such as rainfall, floods, and droughts. Crop failures due to extreme weather conditions are common. For example, wheat is restricted to the winter season, while monsoon rains may damage cotton and rice crops.
Labour and Automation	Largely automated, often using AI-based systems to regulate irrigation, lighting, and climate control. Reduces the need for manual labour, though initial investment in automation can be costly and may limit rural adoption in countries like Pakistan.	Relies heavily on manual labour, especially in rural regions. Limited use of modern equipment and outdated practices reduce efficiency and increase human effort, impacting productivity.

Crop Varieties	Limited to short-cycle, compact crops such as leafy greens, herbs, and select fruits. Not ideal for growing staple grains or large-root crops due to space and weight limitations.	Supports a wide range of crops including cereals (wheat, rice, maize), vegetables (potatoes, onions), and fruits (mangoes, oranges). Thus, it plays a vital role in national food security.
Pesticide and Chemical Use	Controlled environments reduce the risk of pest infestations, minimizing the need for chemical pesticides and herbicides. This leads to safer, chemical-free produce and reduces environmental contamination.	High dependency on pesticides and synthetic fertilizers to manage pests and enhance yields. Overuse may lead to pesticide resistance, health concerns, and environmental degradation through soil and water pollution.
Environmental Impact	Offers a sustainable model with reduced land and water usage, lower CO ₂ emissions, and no soil erosion. Often located near consumers, reducing transport-related emissions. However, energy demands (e.g., for LED lights) may be offset using renewable sources like solar power.	Faces environmental threats such as land degradation, deforestation, waterlogging, and salinization. Expansion often leads to loss of biodiversity and soil fertility. Traditional agriculture is vulnerable to climate change and extreme weather conditions.
Accessibility and Investment	Requires significant initial investment for infrastructure, climate regulation, and automation systems. Thus, it is currently more accessible to commercial enterprises in urban settings than to small-scale rural farmers.	Involves relatively lower startup costs and is more accessible to small-scale farmers using basic tools and natural resources. However, productivity is generally lower and more susceptible to environmental variability.

SLOs Based (MCQs)

VERTICAL FARMING

- Which of the following is a key characteristic of vertical farming?
 - Use of synthetic fertilizers
 - Cultivation in arid lands
 - Use of soil in multi-level platforms
 - Cultivation in vertically stacked layers
- What is the main reason vertical farming avoids using soil?
 - Soil cannot retain nutrients
 - Soil attracts pests
 - Soil is too heavy for vertical structures
 - Soil lacks oxygen
- In hydroponics, plants receive nutrients through:
 - Suspended air mists
 - Decomposing organic matter
 - Soil absorption
 - Circulating nutrient-rich water
- Which vertical farming method provides the highest oxygen availability to plant roots?
 - Hydroponics
 - Soil-based culture
 - Aeroponics
 - Aquaponics
- Aquaponics combines hydroponics with:
 - Composting
 - Livestock farming
 - Aquaculture
 - Mushroom farming
- What serves as a natural fertilizer in aquaponics systems?
 - Vermicompost
 - Fish waste
 - Seaweed extract
 - Urea
- Vertical farms achieve water savings of up to:
 - 40%
 - 60%
 - 90%
 - 10%
- Which technology is used in Controlled Environment Agriculture (CEA)?
 - Pesticide drones
 - Solar heaters
 - LED grow lights
 - Organic mulch
- The major advantage of vertical farming in urban areas is:
 - Cheaper land availability
 - Easier rainfall access
 - Efficient use of limited space
 - Crop rotation options



10. Vertical farming is more sustainable than traditional farming because it:
 - A. Increases pesticide use
 - B. Expands farmland
 - C. Minimizes land and water use
 - D. Promotes seasonal production
11. The type of vertical farming that uses misted nutrient water is:
 - A. Aquaponics
 - B. Hydroponics
 - C. Soil-based farming
 - D. Aeroponics
12. Traditional agriculture in Pakistan primarily relies on:
 - A. AI-based automation
 - B. Indoor growing chambers
 - C. Manual labor and irrigation
 - D. Drone-based planting
13. Which crop is least suitable for vertical farming?
 - A. Spinach
 - B. Strawberries
 - C. Wheat
 - D. Basil
14. A major limitation of vertical farming in Pakistan is:
 - A. Lack of water
 - B. High energy costs
 - C. Soil salinization
 - D. Inaccessibility to remote regions
15. Which crop season in Pakistan includes rice and cotton?
 - A. Summer
 - B. Rabi
 - C. Kharif
 - D. Monsoon
16. A primary environmental concern in traditional agriculture is:
 - A. Indoor air pollution
 - B. Biodiversity loss and soil degradation
 - C. Over-fertilization of hydroponics
 - D. Lack of pesticide use
17. Vertical farming contributes to food security by:
 - A. Growing staple cereals efficiently
 - B. Expanding arable land
 - C. Providing reliable year-round yields
 - D. Increasing livestock productivity
18. Which Pakistani provinces are noted for extensive irrigated land use in traditional farming?
 - A. Khyber Pakhtunkhwa and Balochistan
 - B. Punjab and Sindh
 - C. Azad Kashmir and Gilgit-Baltistan
 - D. Islamabad and Rawalpindi
19. The role of AI in vertical farming includes:
 - A. Transporting crops
 - B. Monitoring plant health and adjusting environments
 - C. Harvesting grain by hand
 - D. Managing crop rotation
20. What reduces the environmental contamination in vertical farming?
 - A. Overuse of fertilizers
 - B. Limited oxygen exposure
 - C. Controlled pesticide-free environments
 - D. Excessive water usage

Answer Key

1. D	2. C	3. D	4. C	5. C	6. B	7. C	8. C	9. C	10. C
11.D	12.C	13.C	14.B	15.C	16.B	17.C	18.B	19.B	20.C

Short Questions

Q.1 What is vertical farming and where is it commonly established?

Ans. Vertical farming is an innovative agricultural method in which crops are grown in vertically stacked layers, often using soil-free systems like hydroponics or aeroponics. This approach maximizes space efficiency, making it ideal for urban environments where arable land is scarce.

Vertical farms can be established in greenhouses, warehouses, skyscrapers, and even indoor spaces such as offices or homes. By utilizing vertical space, controlled lighting, and nutrient delivery systems, vertical farming enables year-round crop production, reduces land use, and minimizes transportation costs by growing food closer to consumers. It is a key strategy for sustainable urban agriculture and addressing food security challenges.

MARKING SCHEME (3 Marks)

- Definition – 2 marks, location – 1 mark

Q.2 How does hydroponics function in vertical farming?

Ans. Hydroponics is a soil-less cultivation technique where plants grow with their roots immersed in a circulating nutrient-rich water solution. This method allows direct absorption of essential minerals and nutrients, ensuring rapid growth and higher yields. Water is continuously

MARKING SCHEME (3 Marks)

- Function – 2 marks, advantages – 1 mark



recycling, conserving up to 90% more water than traditional soil-based farming. Hydroponics reduces dependency on soil quality, minimizes pests and diseases, and enables precise control over nutrient delivery and pH levels. Its efficiency and adaptability make it an ideal method for vertical farming, supporting sustainable and high-density crop production in urban and indoor environments.

Q.3 What is the significance of aeroponics in vertical farming?

Ans. Aeroponics is a cultivation method where plant roots are suspended in the air and periodically misted with nutrient solutions. This approach maximizes oxygen availability to roots, enhances nutrient uptake, and accelerates plant growth. Aeroponics uses minimal water and eliminates soil, reducing risks of soil-borne diseases and pests. It also allows precise control over environmental conditions, supporting higher crop density and yield. These advantages make aeroponics particularly suitable for vertical farming and controlled environment agriculture. By combining efficiency, sustainability, and resource conservation, aeroponics contributes to year-round production of high-quality crops in urban or space-limited settings.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, advantages – 2 marks

Q.4 Explain how aquaponics supports sustainable agriculture.

Ans. Aquaponics integrates hydroponic plant cultivation with aquaculture, creating a closed-loop, sustainable ecosystem. Fish produce nutrient-rich waste, which is converted by bacteria into forms usable by plants. The plants, in turn, filter and purify the water, which is recirculated back to the fish tanks. This system minimizes the need for chemical fertilizers and conserves water through continuous recycling. Aquaponics supports high-density crop and fish production simultaneously, making it highly resource-efficient. By reducing environmental impact and providing both plant and protein yields, aquaponics represents a sustainable and environmentally friendly agricultural approach, especially suited for urban farming, areas with limited water, and controlled environment agriculture systems.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, advantages – 2 marks

Q.5 What technologies support Controlled Environment Agriculture (CEA)?

Ans. Controlled Environment Agriculture (CEA) relies on technologies that optimize growth conditions for crops. Key technologies include LED grow lights that provide tailored light spectra for photosynthesis, climate control systems that maintain temperature and humidity within ideal ranges, and sensor networks that monitor and regulate nutrient levels, pH, and environmental parameters. Automation and data analytics allow precise, consistent, and efficient crop management, supporting year-round production regardless of external weather conditions. These technologies enable higher yields, resource efficiency, and sustainability, making CEA a critical component of modern urban agriculture, vertical farming, and future food security strategies.

MARKING SCHEME (3 Marks)

- 3 names of technologies – 1 mark for three names

Q.6 How does vertical farming conserve water?

Ans. Vertical farming conserves water by using soil-free systems like hydroponics and aeroponics, which deliver nutrient solutions directly to plant roots. Water is continuously recirculated, minimizing losses through evaporation or runoff. Unlike traditional soil-based irrigation, which can lose large amounts of water, vertical farming systems ensure precise control over usage, resulting in up to 95% water savings. This efficiency makes vertical farming particularly suitable for urban environments or regions with water scarcity. By reducing dependency on freshwater resources, vertical farming supports sustainable agriculture while maintaining high crop yields, demonstrating a significant environmental advantage over conventional farming practices.

MARKING SCHEME (3 Marks)

- Advantages – 3 marks

Q.7 Compare crop variety limitations of vertical and traditional farming.

Ans. Vertical farming generally supports compact, fast-growing crops such as leafy greens, herbs, and strawberries, due to limited vertical space and shorter growth cycles. Large or deep-rooted crops like grains, tubers, or tree fruits are difficult to cultivate in stacked or controlled environments. In contrast, traditional farming accommodates a wide range of crops, including cereals, vegetables, and fruits essential for food security, as it is not restricted by vertical space. While vertical farming excels in efficiency and resource use for select crops, traditional farming remains necessary for diverse agricultural production, highlighting a trade-off between yield efficiency and crop variety.

MARKING SCHEME (3 Marks)

- Crop variety limitations – 1.5 marks each for vertical and traditional farming



Q.8 Why is vertical farming considered more sustainable than traditional farming?

Ans. Vertical farming is considered more sustainable because it minimizes land use, reducing deforestation and soil degradation. Soil-free systems prevent erosion, conserve water, and decrease pesticide and fertilizer use. Growing crops in urban or controlled environments reduces transportation emissions and associated carbon footprints. Although vertical farming is energy-intensive, renewable energy sources such as solar panels or LED lighting can offset these demands, making the system environmentally friendly. By combining water efficiency, reduced chemical use, and urban proximity, vertical farming supports resource conservation, high productivity, and lower environmental impact, offering a sustainable alternative to conventional soil-based agriculture, especially in densely populated areas.

MARKING SCHEME (3 Marks)

- Advantages of vertical farming – 3 marks

Q.9 How does automation impact vertical farming compared to traditional farming?

Ans. Automation in vertical farming uses AI, sensor networks, and robotic systems to control irrigation, nutrient delivery, lighting, and climate conditions with precision. This reduces labor requirements, improves efficiency, and ensures consistent crop quality. In contrast, traditional farming, particularly in countries like Pakistan, relies heavily on manual labor, simple tools, and seasonal monitoring, making it less precise and more labor-intensive. Automation allows vertical farms to optimize resources, reduce errors, and maximize yields per unit area, whereas traditional methods may be slower, variable, and more resource-consuming. This technological difference highlights the efficiency and modernity of vertical farming over conventional agricultural practices.

MARKING SCHEME (3 Marks)

- Vertical farming – 1.5 marks,
traditional farming – 1.5 marks

Q.10 What are the limitations of vertical farming in developing countries like Pakistan?

Ans. Vertical farming in developing countries faces several constraints. The high initial cost of infrastructure, including LED lighting, hydroponic systems, and climate control, limits accessibility for small-scale farmers. Energy requirements for artificial lighting and climate management are significant and may not be reliably met in regions with unstable power supply. Additionally, vertical farming is generally restricted to a narrow range of crops, primarily leafy greens and herbs, limiting its ability to replace traditional, staple food production. These factors make vertical farming more feasible for commercial urban operations rather than rural or smallholder farms, slowing its widespread adoption and contribution to local food security.

MARKING SCHEME (3 Marks)

- Three main limitations – 1 mark each

Exercise

I

Multiple Choice Questions (MCQs)

❖ Select the correct Answer.

- Which tool of recombinant DNA technology is incorrectly paired with its use?
A. Restriction enzyme - RFLP production
B. DNA ligase - cuts DNA
C. Reverse transcriptase - makes cDNA
D. Electrophoresis - separates DNA
- Which of the following is a key difference between vertical farming and traditional agriculture?
A. Vertical farming uses more horizontal space, suitable for rural areas.
B. Vertical farming is not ideal for crops like wheat, rice, and sugarcane.
C. Vertical farming is ideal for urban areas; traditional farming needs large fields.
D. Traditional farming grows crops in vertical layers; vertical farming requires vast fields.
- A paleontologist finds tissue from a 400-year-old extinct bird and wants to compare its DNA with living birds. Which method would best increase the bird DNA for testing?
A. a RFLP analysis
B. polymerase chain reaction (PCR)
C. electroporation
D. Southern hybridization
- Which of the following sequences in double-stranded DNA is mostly likely to be recognized as a cutting site for a restriction enzyme?



- A. AAGGTTCC
C. GGCCCCGG
5. **A plasmid**
A. is used as DNA vector
B. is a type of bacteriophage
C. is a type of CDNA
D. is a retrovirus
6. **DNA molecules with complementary sticky ends associate by**
A. covalent bond
C. ionic bond
7. **Which technique rapidly replicates specific DNA fragments without cloning?**
A. gel electrophoresis
C. genetic probe
D. polymerase chain reaction
8. **The PCR technique uses**

B. AGTCTCAG
D. ACCATGGT

- A. heat resistant DNA polymerase
B. reverse transcriptase
C. DNA ligase
D. restriction enzymes
9. **If eukaryotic DNA contains five cleavage sites for a particular restriction enzyme, how many fragments will be produced upon complete digestion of the DNA with that enzyme?**
A. 2
C. 6
10. **The dideoxynucleotides (ddATP, ddTTP, ddGTP and ddCTP) are important in DNA sequencing because they:**
A. cause premature termination of growing DNA strand
B. are used as prime
C. cause the DNA fragments that contain them to migrate more slowly through the gel
D. are not affected by high temperatures

Answer Key

1. B	2. C	3. B	4. D	5. A	6. B	7. D	8. A	9. C	10. A
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II

Short Answer Questions

Q.1 What is the role of restriction endonucleases in gene cloning?

Ans. Restriction endonucleases are enzymes that recognize specific palindromic DNA sequences and cleave the DNA at or near these sites. In gene cloning, they are essential for creating recombinant DNA, as they cut both the gene of interest and the vector DNA at complementary sequences. This generates sticky or blunt ends that allow precise insertion of the target gene into the vector. By producing predictable fragments, restriction endonucleases enable researchers to manipulate DNA with high specificity, facilitating cloning, sequencing, and genetic engineering. Their ability to target specific sequences makes them fundamental tools in molecular biology and biotechnology for applications such as therapeutic protein production, GMO development, and genetic research.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, role in cloning – 2 marks

Q.2 What are molecular carriers?

Ans. Molecular carriers, also known as vectors, are DNA molecules that transport foreign genetic material into a host cell for replication or expression. Common vectors include plasmids, bacteriophages, cosmids, and artificial chromosomes. Effective vectors possess features such as an origin of replication to enable self-replication within the host, selectable marker genes for identification of transformed cells, and multiple cloning sites to facilitate insertion of target genes. Vectors are crucial in gene cloning, recombinant protein production, and genetic engineering, allowing scientists to introduce and propagate specific DNA sequences in bacterial, yeast, or mammalian systems, forming the foundation of modern biotechnology and molecular medicine.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, examples and role – 2 marks

Q.3 What is the role of restriction DNA ligases in gene technology?

Ans. DNA ligases are enzymes that catalyze the formation of phosphodiester bonds between adjacent nucleotides in DNA strands. In gene technology, they are used to covalently join DNA fragments, such as linking a gene of interest to a plasmid vector to form recombinant DNA.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, role – 2 marks



By sealing the sugar-phosphate backbone, DNA ligases ensure stable and continuous DNA molecules that can be replicated and expressed in host organisms. They are vital for molecular cloning, genetic engineering, and repair of DNA during laboratory manipulations. Without DNA ligases, the assembly of functional recombinant DNA constructs for protein expression or gene therapy would not be possible.

Q.4 What are the concerns about genetically modified organisms (GMOs)?

Ans. Concerns about GMOs include potential ecological disruption, such as unintended effects on non-target species or loss of biodiversity. There is also the risk of horizontal gene transfer, where antibiotic resistance markers could spread to pathogens. Food safety issues like allergenicity or toxicity are debated, alongside ethical questions about manipulating life forms. Socioeconomic concerns include patent control over genetically engineered seeds by large biotech companies, which may affect farmers' independence and food accessibility. Additionally, there is public apprehension regarding long-term health and environmental impacts. These concerns highlight the importance of strict regulation, biosafety assessment, and ethical oversight in GMO research and commercialization.

MARKING SCHEME (3 Marks)

- 3 main concerns – 1 mark each

Q.5 Define following terms:

(i) Biotechnology

Ans. Biotechnology is the application of biological systems, organisms, or their components to develop useful products or processes for specific purposes. It spans multiple fields, including medicine, agriculture, industry, and environmental management. In medicine, biotechnology enables the production of vaccines, therapeutic proteins, and diagnostic tools. In agriculture, it helps develop pest-resistant or nutrient-enhanced crops. Industrially, it is applied in biofuels, fermentation, and bioprocessing. Environmental applications include bioremediation and conservation of biodiversity. By leveraging the capabilities of living organisms, biotechnology integrates scientific knowledge with practical solutions, improving human welfare, food security, and sustainable development while offering innovative approaches to solve complex biological and ecological challenges.

MARKING SCHEME (2 Marks)

- Definition – 2 marks

(ii) Genetic Engineering

Ans. Genetic engineering is the deliberate modification of an organism's genome using biotechnological techniques. This includes the insertion, deletion, or alteration of genes to change genetic characteristics. It enables scientists to produce organisms with desirable traits, such as disease resistance in plants, production of human proteins in bacteria, or enhanced growth in animals. Genetic engineering relies on tools such as recombinant DNA technology, gene cloning, and CRISPR-Cas9 for precise genome editing. Applications range from agriculture and medicine to industrial biotechnology. The field also raises ethical and safety considerations, requiring careful regulation. Overall, genetic engineering is a cornerstone of modern biotechnology, allowing controlled manipulation of life at the molecular level.

MARKING SCHEME (2 Marks)

- Definition – 2 marks

(iii) Gene Cloning.

Ans. Gene cloning is the process of producing multiple identical copies of a specific gene or DNA segment. The gene of interest is inserted into a suitable vector, such as a plasmid, which is then introduced into a host organism, typically bacteria. Within the host, the recombinant DNA is replicated, generating numerous copies of the target gene. Gene cloning allows scientists to study genes in detail, produce recombinant proteins, and perform genetic engineering experiments. It is fundamental for applications such as insulin production, vaccine development, and functional genomics. By enabling precise replication of DNA sequences, gene cloning serves as a foundational tool in molecular biology and biotechnology.

MARKING SCHEME (2 Marks)

- Definition – 2 marks

(iv) Palindromic Sequence in DNA.

Ans. A palindromic sequence in DNA is a nucleotide sequence that reads the same forward on one strand and backward on the complementary strand. For example, GAATTC on one strand pairs with CTTAAG on the other. These sequences are recognized by restriction enzymes, which cleave DNA at or near these sites, producing sticky or blunt ends for recombinant DNA construction.

MARKING SCHEME (2 Marks)

- Definition – 2 marks



Palindromic sequences are essential in gene cloning, molecular diagnostics, and genetic engineering because they allow precise targeting and manipulation of DNA fragments. By providing specificity, these sequences facilitate the insertion, deletion, or modification of genes in biotechnology applications.

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(v) Recombinant DNA Technology

Ans. Recombinant DNA technology involves combining DNA molecules from different sources to create a new DNA sequence. This is achieved by inserting a gene of interest into a vector, such as a plasmid, and introducing it into a host organism, where it can be replicated and expressed. The resulting recombinant DNA can produce proteins, enzymes, or other biomolecules for medical, agricultural, or industrial purposes. Applications include insulin production, genetically modified crops, and gene therapy. Recombinant DNA technology relies on molecular tools like restriction enzymes, DNA ligases, and vectors, and it forms the foundation of modern genetic engineering, enabling targeted manipulation of genetic material to achieve desired outcomes.

MARKING SCHEME (2 Marks)

- Definition – 2 marks

(vi) DNA Ligase

Ans. DNA ligase is an enzyme that catalyzes the formation of phosphodiester bonds between adjacent nucleotides, effectively joining two DNA fragments. It plays a vital role in DNA replication and repair by sealing nicks in the sugar-phosphate backbone. In molecular biology, DNA ligase is essential for recombinant DNA technology, where it covalently links a gene of interest into a vector to produce stable recombinant molecules. This enables gene cloning, protein expression, and genetic engineering experiments. By ensuring the continuity and integrity of DNA strands, DNA ligase facilitates accurate genetic manipulation and is fundamental to modern biotechnology applications in medicine, agriculture, and research.

MARKING SCHEME (2 Marks)

- Definition – 2 marks

(vii) Polymerase Chain Reaction (PCR)

Ans. PCR is an in vitro molecular technique used to amplify a specific DNA segment exponentially. It involves repeated cycles of denaturation, where the DNA strands separate; annealing, where short primers bind to complementary sequences flanking the target region; and extension, where a thermostable DNA polymerase synthesizes new strands. PCR allows the production of millions of copies of a DNA fragment from minimal starting material in a short time. It is widely used in diagnostics, forensic analysis, gene cloning, and research. By enabling rapid and precise amplification of DNA, PCR has become an indispensable tool in modern biotechnology and molecular biology.

MARKING SCHEME (2 Marks)

- Definition – 2 marks

(viii) Primers in the context of PCR

Ans. Primers are short, single-stranded DNA sequences designed to bind specifically to complementary regions flanking the DNA segment to be amplified in PCR. They provide a starting point for DNA polymerase to initiate synthesis of new DNA strands. Forward and reverse primers define the boundaries of the target sequence and ensure specificity. Accurate primer design is critical for efficient amplification and avoiding non-specific products. Primers are fundamental in PCR applications, including diagnostics, gene cloning, sequencing, and detection of genetic mutations. They enable selective amplification of desired DNA fragments, making molecular experiments precise, reproducible, and reliable.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, function – 2 marks

(ix) Taq Polymerase

Ans. Taq polymerase is a thermostable DNA polymerase extracted from the bacterium *Thermus aquaticus*. It synthesizes new DNA strands during the extension step of PCR and can withstand repeated cycles of high temperature required for DNA denaturation without losing activity. Its heat stability makes it ideal for PCR, enabling efficient and rapid amplification of DNA fragments. Taq polymerase is widely used in molecular biology for cloning, diagnostics, sequencing, and research applications. Its discovery revolutionized PCR technology, making it reliable, fast, and accessible for laboratories worldwide.

MARKING SCHEME (3 Marks)

- Definition – 1 mark, importance in PCR – 2 marks



(x) Genome

Ans. A genome is the complete set of genetic material present in an organism, including all coding genes, regulatory sequences, and non-coding DNA, organized into chromosomes. It contains the instructions necessary for growth, development, and reproduction. Genomes vary in size and complexity across species, from small viral genomes to large eukaryotic genomes. Genome analysis allows identification of genes, regulatory networks, and evolutionary relationships. Mapping and sequencing genomes underpin modern genetics, biotechnology, and medicine, facilitating advances in gene therapy, personalized medicine, crop improvement, and conservation biology. Understanding genomes is essential for decoding biological information and applying it in research and applied sciences.

MARKING SCHEME (2 Marks)

- Definition – 2 marks

28
28

(xi) Transgenic Organisms

Ans. Transgenic organisms are those that carry one or more genes artificially inserted from another species through recombinant DNA technology. These introduced genes confer new traits or functions, such as pest resistance in plants, enhanced growth in animals, or production of therapeutic proteins in bacteria. Creating transgenic organisms involves identifying a gene of interest, inserting it into a vector, and introducing it into a host organism where it can be stably expressed. Transgenic organisms have applications in medicine, agriculture, and industry, including genetically modified crops, protein production, and animal models for disease research. They are a cornerstone of modern biotechnology, offering innovative solutions to human challenges.

MARKING SCHEME (2 Marks)

- Definition – 2 marks

Q.6 Write the difference between:

(a) Biotechnology and Genetic Engineering

Ans. Biotechnology is a broad field that uses living organisms or their components to develop products and processes across medicine, agriculture, industry, and the environment. It includes both modern molecular techniques and traditional practices such as fermentation and selective breeding. Genetic engineering, on the other hand, is a specialized branch of biotechnology that focuses on direct manipulation of genes through techniques like recombinant DNA technology and gene editing. While biotechnology encompasses a wide range of applications, genetic engineering is primarily used for modifying specific genetic traits in organisms, producing transgenic crops, therapeutic proteins, or animal models for research. Genetic engineering is thus a precise, gene-level tool within the broader scope of biotechnology.

MARKING SCHEME (3 Marks)

- Biotechnology – 1.5 marks, genetic engineering – 1.5 marks

(b) Staggered cut and Blunt cut Restriction Enzymes

Ans. Staggered cut restriction enzymes cleave DNA to produce short, single-stranded overhangs known as sticky ends, which facilitate easier and more specific joining of DNA fragments during recombinant DNA experiments. An example is EcoRI. Blunt cut restriction enzymes cut straight across both DNA strands without creating overhangs, making ligation less efficient; SmaI is an example. Sticky ends enhance the efficiency and specificity of DNA fragment joining, while blunt ends require more careful handling during ligation. Both types are essential tools in molecular cloning, genetic engineering, and recombinant DNA technology, allowing scientists to manipulate and assemble DNA fragments for gene cloning, transgenic development, and other biotechnological applications.

MARKING SCHEME (3 Marks)

- Staggered cut – 1.5 marks, blunt cut – 1.5 marks

(c) Genome and Chromosome

Ans. A genome is the complete set of genetic material in an organism, including all coding genes, non-coding sequences, and regulatory elements. A chromosome is a single, organized DNA molecule within the nucleus that carries specific genes. The human genome consists of 46 chromosomes (23 pairs), each forming a part of the overall genome. While the genome represents the totality of an organism's hereditary information, chromosomes are the physical structures that package and organize this DNA for replication, transcription, and inheritance. Understanding both concepts is crucial for genetics, genomics, and biotechnology, as they provide insight into how genes are organized, regulated, and expressed in living organisms.

MARKING SCHEME (3 Marks)

- Genome – 1.5 marks, chromosome – 1.5 marks



(d) Transgenic Organisms and Hybrid Organisms

Ans. Transgenic organisms contain foreign genes artificially inserted through genetic engineering, giving them traits not naturally present in the species. For example, Bt cotton carries an insect-resistance gene from bacteria. **Hybrid organisms** result from sexual cross-breeding of two different species or varieties, combining traits from both parents, such as a mule, produced from a horse and a donkey. Transgenic organisms are created using molecular biology techniques, while hybrids result from traditional breeding methods. Both approaches are used to enhance desirable traits, but transgenic organisms allow for precise gene-level modifications, whereas hybrids are limited to naturally compatible species or varieties.

MARKING SCHEME (3 Marks)

- Transgenic organisms – 1.5 marks, hybrid organisms – 1.5 marks

(e) Biotechnology and Nanotechnology

Ans. **Biotechnology** manipulates biological systems, such as cells, DNA, enzymes, and organisms, to produce applications in medicine, agriculture, and industry, including genetic engineering, biopharmaceuticals, and biofuels.

MARKING SCHEME (3 Marks)

- Biotechnology – 1.5 marks, nanotechnology – 1.5 marks

Nanotechnology, in contrast, works at the nanoscale (1–100 nm) with atoms, molecules, and nanoparticles to create materials and devices with novel properties, such as nanosensors, targeted drug delivery, or nanomaterials. Biotechnology focuses on living systems, while nanotechnology focuses on matter at the atomic or molecular level. Both fields have transformative impacts on science and industry, but their tools, targets, and applications differ fundamentally, with biotechnology emphasizing biological processes and nanotechnology emphasizing physical and chemical manipulation at extremely small scales.

Q.7 Shortly suggest, which type of farming is better from vertical farming and traditional agriculture farming in Pakistan.

Ans. Traditional agriculture remains more practical in Pakistan due to extensive rural areas, established farming infrastructure, and staple crops like wheat and rice, which are not suited for vertical farming systems. Traditional farming supports large-scale cereal and pulse production, critical for food security. However, vertical farming offers advantages for urban areas, including high-value crops such as leafy greens, herbs, and strawberries, maximizing space and water efficiency. It also reduces dependency on arable land and mitigates the impact of climate variability. While vertical farming has potential for supplementing food production in cities, Pakistan's agricultural economy currently relies predominantly on traditional soil-based farming for widespread staple crop cultivation.

MARKING SCHEME (3 Marks)

- Comparison – 1.5 marks each for vertical and traditional farming

Q.8 Explain the role of restriction endonucleases and DNA ligases in gene cloning.

Ans. Restriction endonucleases are enzymes that recognize specific palindromic DNA sequences and cleave DNA at those sites, producing sticky or blunt ends. This allows the precise isolation of a gene of interest and preparation of a vector for insertion. DNA ligases are enzymes that join these DNA fragments by forming covalent phosphodiester bonds, creating stable recombinant DNA molecules. Together, these enzymes enable the construction of recombinant DNA, which can be replicated and expressed in a host organism. Their combined function is fundamental to molecular cloning, gene manipulation, and biotechnology, ensuring specificity, stability, and efficiency in producing genetically engineered organisms or proteins.

MARKING SCHEME (3 Marks)

- Restriction endonucleases – 1 mark, DNA ligases – 2 marks

Q.9 Describe the selection and isolation of the gene of interest.

Ans. The process begins with identifying the required gene from genomic DNA or complementary DNA (cDNA). Restriction enzymes are then applied to cut the DNA at specific palindromic sites. This cutting produces sticky or blunt ends, which help in separating the target gene. In this way, the desired gene is isolated and prepared for cloning.

MARKING SCHEME (3 Marks)

- Identification of gene from genomic DNA or cDNA (1 mark)
- Cutting by restriction enzymes at specific sites (1 mark)
- Formation of sticky/blunt ends for isolation (1 mark)

Q.10 Explain the properties and the role of vectors in recombinant DNA technology.

Ans. Vectors are DNA molecules used to carry foreign genes into host cells. Ideal vectors possess an origin of replication to allow independent replication, selectable marker genes (e.g., antibiotic resistance) for

MARKING SCHEME (3 Marks)

- Properties – 1.5 marks, role – 1.5 marks



identifying transformed cells, and multiple cloning sites to facilitate insertion of the target gene. Vectors can be plasmids, bacteriophages, cosmids, or artificial chromosomes. Their role is to transport, replicate, and enable expression of the gene of interest within a host organism, forming recombinant DNA. Vectors are indispensable in molecular cloning, gene expression studies, protein production, and genetic engineering, providing stability, specificity, and efficiency in introducing genetic material into biological systems.

Q.11 State the steps for the integration of DNA insert into the vector.

Ans. The integration of a DNA insert into a vector involves several steps.

First, restriction enzymes cut both the vector and the gene of interest at specific sequences, generating complementary sticky or blunt ends. The DNA insert is then mixed with the vector to allow base pairing of these ends. DNA ligase seals the sugar-phosphate backbone, creating a stable recombinant DNA molecule. Finally, the recombinant plasmid is introduced into a host cell by transformation, where it can replicate and express the inserted gene. This process forms the foundation of recombinant DNA technology, enabling gene cloning, protein production, and various applications in medicine and biotechnology.

MARKING SCHEME (3 Marks)

- 1 mark each for four main steps

Q.12 Briefly state the technique applied for the selection of vectors that take up the DNA insert.

Ans. Vectors that successfully take up DNA inserts are selected using selectable marker genes, such as antibiotic resistance. After transformation, only host cells containing vectors with the inserted

gene survive when cultured in the presence of the corresponding antibiotic. Molecular probes, reporter genes, or PCR amplification may be used to confirm the presence and correct orientation of the insert. This technique ensures that only recombinant vectors are propagated, allowing researchers to isolate and study the gene of interest efficiently. Selection minimizes experimental errors, increases cloning success, and is a critical step in recombinant DNA technology for producing proteins, functional studies, or genetic modifications.

MARKING SCHEME (3 Marks)

- Method of selection – 3 marks

Q.13 Describe the steps involved in gene amplification through polymerase chain reaction (PCR).

Ans. PCR amplifies a specific DNA segment through repeated cycles. First, denaturation heats the DNA to $\sim 95^{\circ}\text{C}$, separating the two strands. Second, annealing lowers the temperature, allowing primers to bind to target sequences on each strand. Third, extension occurs at $\sim 72^{\circ}\text{C}$, where Taq polymerase synthesizes new DNA strands from the primers.

These steps are repeated 25–35 times, resulting in exponential amplification of the target DNA. PCR is rapid, sensitive, and specific, enabling researchers to produce millions of copies of DNA from a small sample. It is widely used in diagnostics, forensics, cloning, and genetic research, forming a cornerstone of modern molecular biology.

MARKING SCHEME (3 Marks)

- 1 mark for each step – denaturation, annealing, extension, and 1 mark for exponential amplification

Q.14 State the objectives of producing transgenic bacteria, transgenic plants, and transgenic animals.

Ans. Transgenic organisms are engineered for specific purposes. Transgenic bacteria produce human proteins, such as insulin, degrade environmental pollutants, or enhance nutrient availability in agriculture. Transgenic plants are developed to resist pests, herbicides,

or drought, and to improve nutritional quality, contributing to food security. Transgenic animals are created to improve growth rates, disease resistance, or to produce pharmaceutical proteins in milk or blood, a process known as transpharming. The overarching objective is to harness genetic engineering to solve human problems, increase productivity, and improve health, while maintaining safety and sustainability in agricultural and medical applications.

MARKING SCHEME (3 Marks)

- 1 mark for each organism – bacteria, plants, animals

Q.15 Explain the scope and importance of biotechnology in promoting human welfare.

Ans. Biotechnology has far-reaching impacts on health, agriculture, and industry. In healthcare, it enables production of life-saving drugs, vaccines, and diagnostics for early disease detection. In agriculture, biotechnology improves crop yield, pest resistance, and nutritional quality, contributing to food security. Environmental applications include bioremediation to clean pollutants

and conservation of biodiversity. Industrial uses include biofuels, biodegradable materials, and fermentation processes. Overall, biotechnology enhances human welfare by improving quality of life, ensuring sustainable

MARKING SCHEME (3 Marks)

- 3 marks for health, agriculture, industry/environment



food production, and addressing environmental and medical challenges. Its scope continues to expand with advances in genetic engineering, molecular biology, and synthetic biology.

Q.16 List the hazards of using genetically modified organisms (GMOs).

Ans. GMOs pose several potential hazards. They may cause allergic reactions or unknown health effects in humans. Gene transfer to non-target species or wild populations can disrupt ecosystems. GMOs may contribute to loss of biodiversity and unforeseen environmental impacts. Ethical concerns arise regarding animal welfare and manipulation of life forms. Additionally, pests or weeds may develop resistance to GMO-based interventions, reducing long-term effectiveness. These hazards highlight the need for rigorous risk assessment, regulatory oversight, and responsible management to ensure that the benefits of GMOs outweigh potential risks for health, agriculture, and the environment.

MARKING SCHEME (3 Marks)
0.5 mark for each main hazards

III

Long Answer Questions

Q.1 Describe the components of recombinant DNA technology under the following heads:

- (a) Gene of interest (b) Molecular scissors (c) Molecular carriers
(d) Molecular glue (e) Expression system

Ans. Please See SLO [B-12-J-01]

Q.2 Describe the mechanism or procedure of recombinant DNA technology.

Ans. Please See SLO [B-12-J-01]

Q.3 What is polymerase chain reaction? Discuss its components and mechanism?

Ans. Please See SLO [B-12-J-02]

Q.4 Describe vertical agriculture farming. Explain hydroponics, aeroponics and aquaponics.

Ans. Please See SLO [B-12-J-07]

Q.5 Discuss the scope and importance of biotechnology in promoting human welfare.

Ans. Please See SLO [B-12-J-05]

Q.6 What are the ethical, legal and social implications of using biotechnology?

Ans. Please See SLO [B-12-J-05]

Q.7 Compare vertical farming and traditional agriculture farming in Pakistan to clear advantages and disadvantages of both.

Ans. Please See SLO [B-12-J-08]

